



News from IU: 5-Star Linked Open Data for Cell Type Populations and Digital Objects from the Human Reference Atlas via the HRA Knowledge Graph



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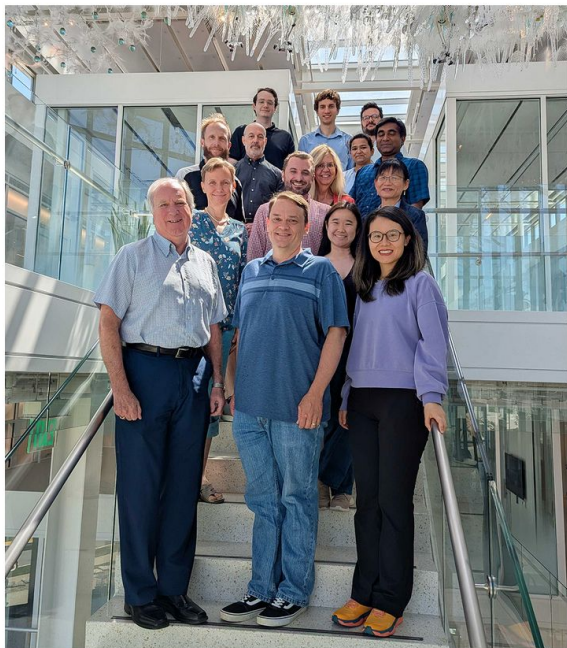
Bruce W. Herr II
Technical Director

*Cyberinfrastructure for Network Science
Center
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Engineering
Indiana University, Bloomington, IN, USA*

HRA Hackathons

Participating Teams

Experts from government, academic, and research institutions came together to share knowledge, tools, and data in support of the Human Reference Atlas initiative.



1st row: Richard Scheuermann, NLM; Bruce W. Herr II, CNS; Yun "Renee" Zhang, NLM.
2nd row: Katy Börner, CNS; Matthew Diller, NLM; Beverly Peng, J. Craig Venter Institute; Bingfang Xu, NLM.
3rd row: Andreas Bueckle, CNS; Raymond Leclair, NLM; Anne Deslattes Mays, NLM; Ajith Pankajam, NLM.
4th row: William Spear, NLM; Noam Rotenberg, NLM; Supriya Bidanta, CNS; Michael Ginda, CNS.

Participants



KPMP Team

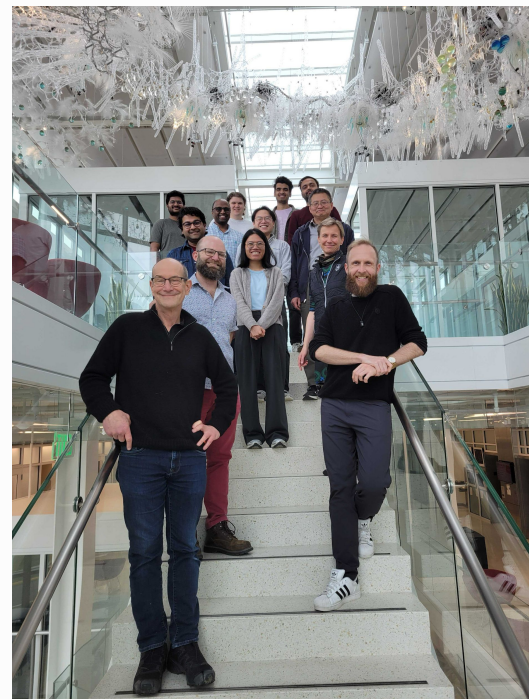
- Yongqun Oliver He, Prof, University of Michigan
- Jie Zheng, Staff, University of Michigan
- Ruopeng Wu, Student, University of Michigan
- Yichao Chen, Student, University of Michigan

HPO Team

- Aybuge Altay, Postdoc, Berlin Institute of Health at Charité (BIH) Center of Health Data Sciences

Indiana University Team

- Katy Börner, CNS and HRA Lead
- Bruce W. Herr II, System Architect
- Andreas Bueckle, Research Lead and HRApop Lead
- Michael Ginda, NLP & HRAlit Expert



Attendees

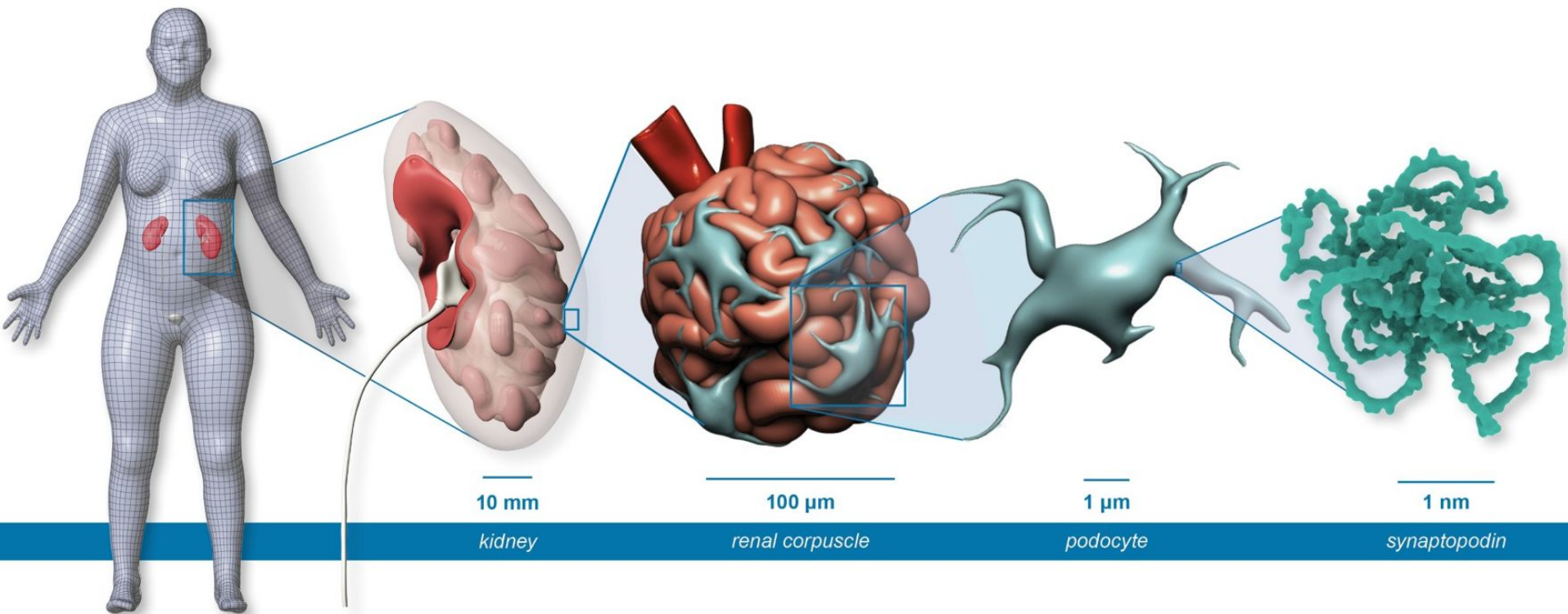
- Eran Agmon, PhD, Assistant Professor, UConn Health ([website](#))
- Bruce J. Aronow, PhD, Professor, University of Cincinnati ([website](#))
- Surya Prasath, PhD, Professor, University of Cincinnati ([website](#))
- Feng-Yu (Leo) Yeh, MS, Software Developer and Bioinformatician, University of Michigan (Oliver He's lab) ([website](#))
- Michael Gargano, Senior Scientific Software Engineer, The Jackson Laboratory ([website](#))
- Daniel Danis, PhD, Research Scientist, Berlin Institute of Health at Charité ([website](#))
- Rupesh Galal, Graduate Student, University of Cincinnati
- Pankaj Rajdeo, Graduate Student, University of Cincinnati
- Daniel Bolin, Senior Software Developer, Indiana University ([website](#))
- Yashvardhan Jain, MS, Research Software Engineer, Indiana University ([website](#))
- Yongxin (Kiki) Kong, PhD, Postdoctoral Fellow, Indiana University
- Bruce W. Herr II, Technical Director, Indiana University ([website](#))
- Katy Börner, PhD, Center Director, Indiana University ([website](#))
- Andreas Bueckle, PhD, Research Lead, Indiana University ([website](#))

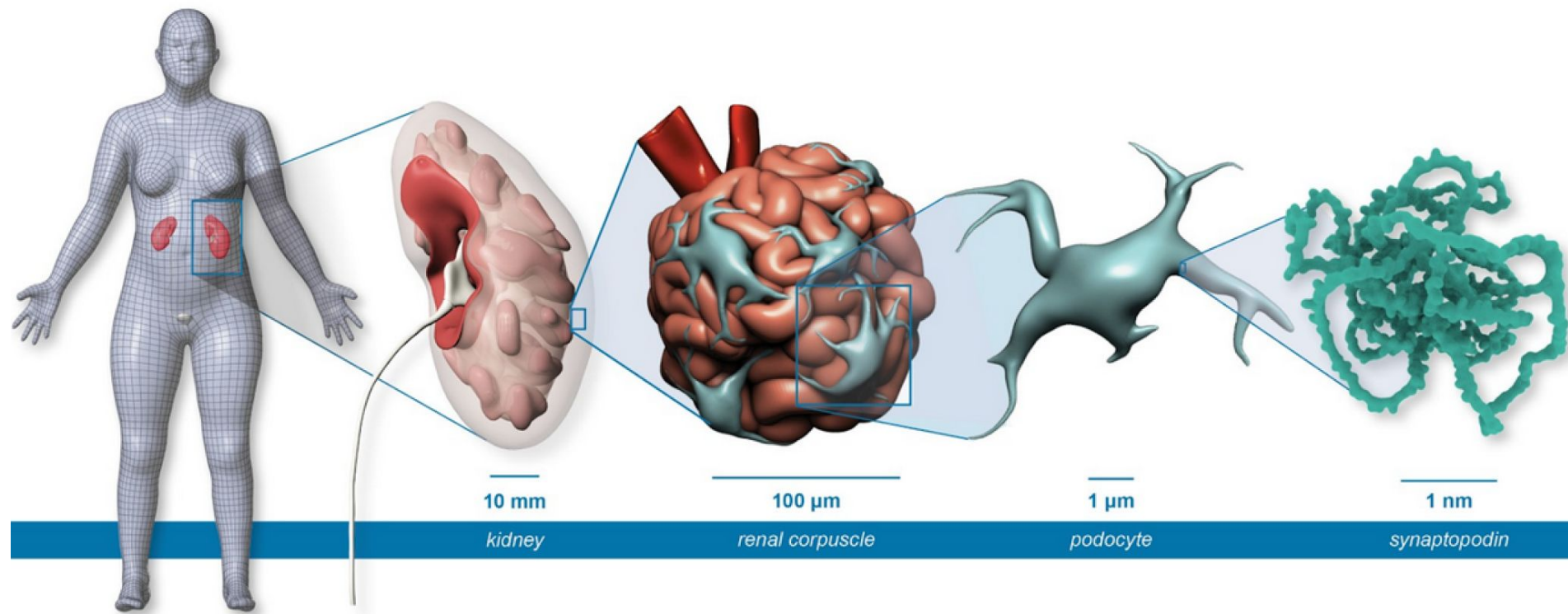


Welcome & Overview: A Tour of the HRA Knowledge Graph (KG)

Human Reference Atlas (HRA)

A comprehensive, ontologically aligned, high-resolution, three-dimensional, multiscale atlas of anatomical structures and cells in the healthy human body



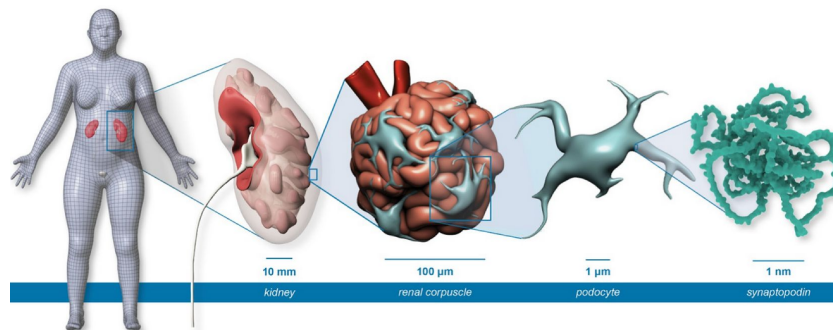


Anatomical Structures

Functional
Tissue Units

Cell Types

Biomarkers
Genes, Proteins, ..



Anatomical Structures

Functional
Tissue Units

Cell Types

Biomarkers
Genes, Proteins, ..

Biology

Anatomical Structures, Cell Types, and Biomarkers Tables

Atlas

3D Reference Organs

2D FTU
Illustrations

Organ Mapping Antibody Panels

Vascular Geometry

Cell Type
Annotations



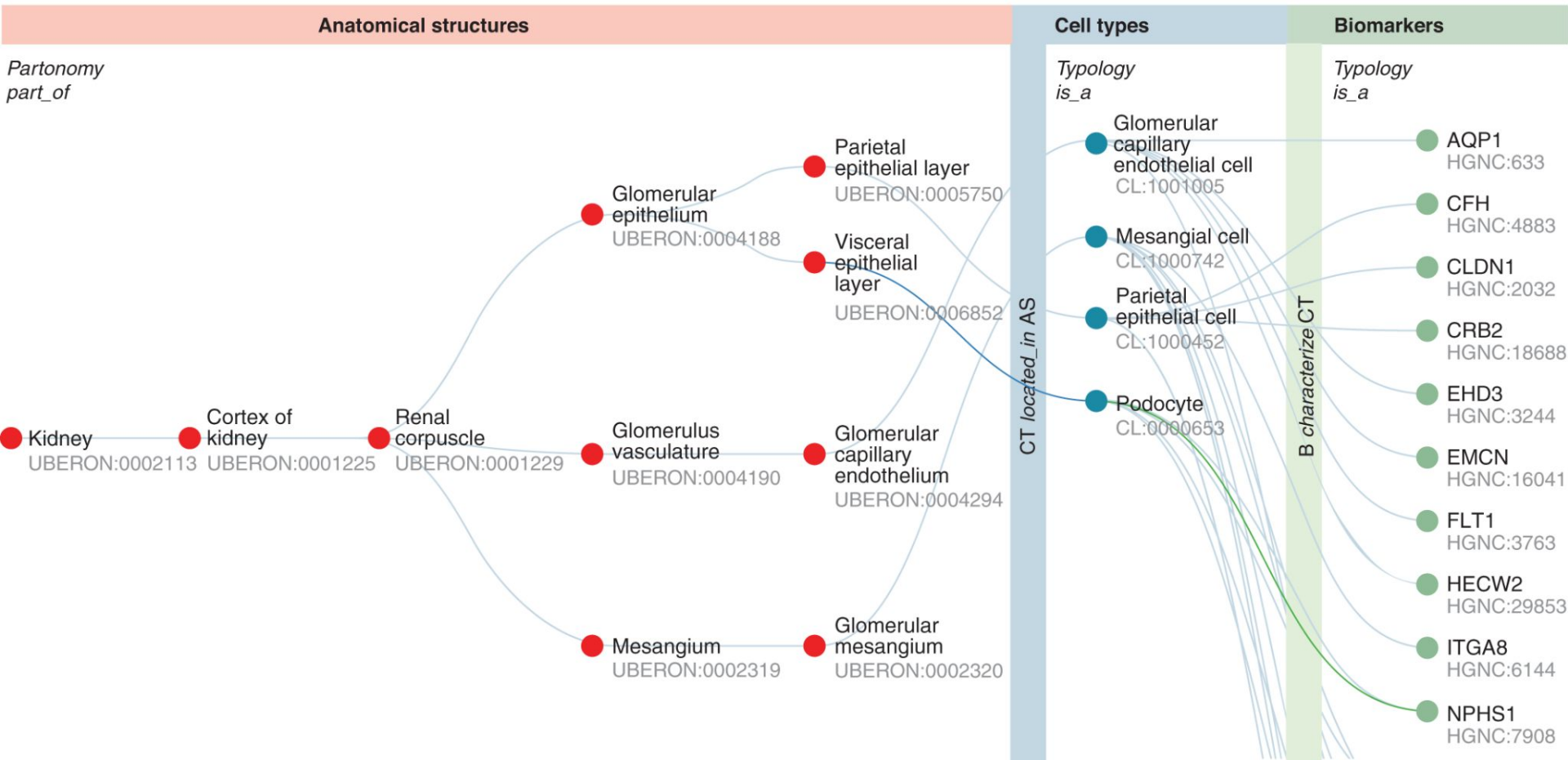
Experimental Data (Donors, Tissues,
Extraction Sites, Datasets)

Atlas++

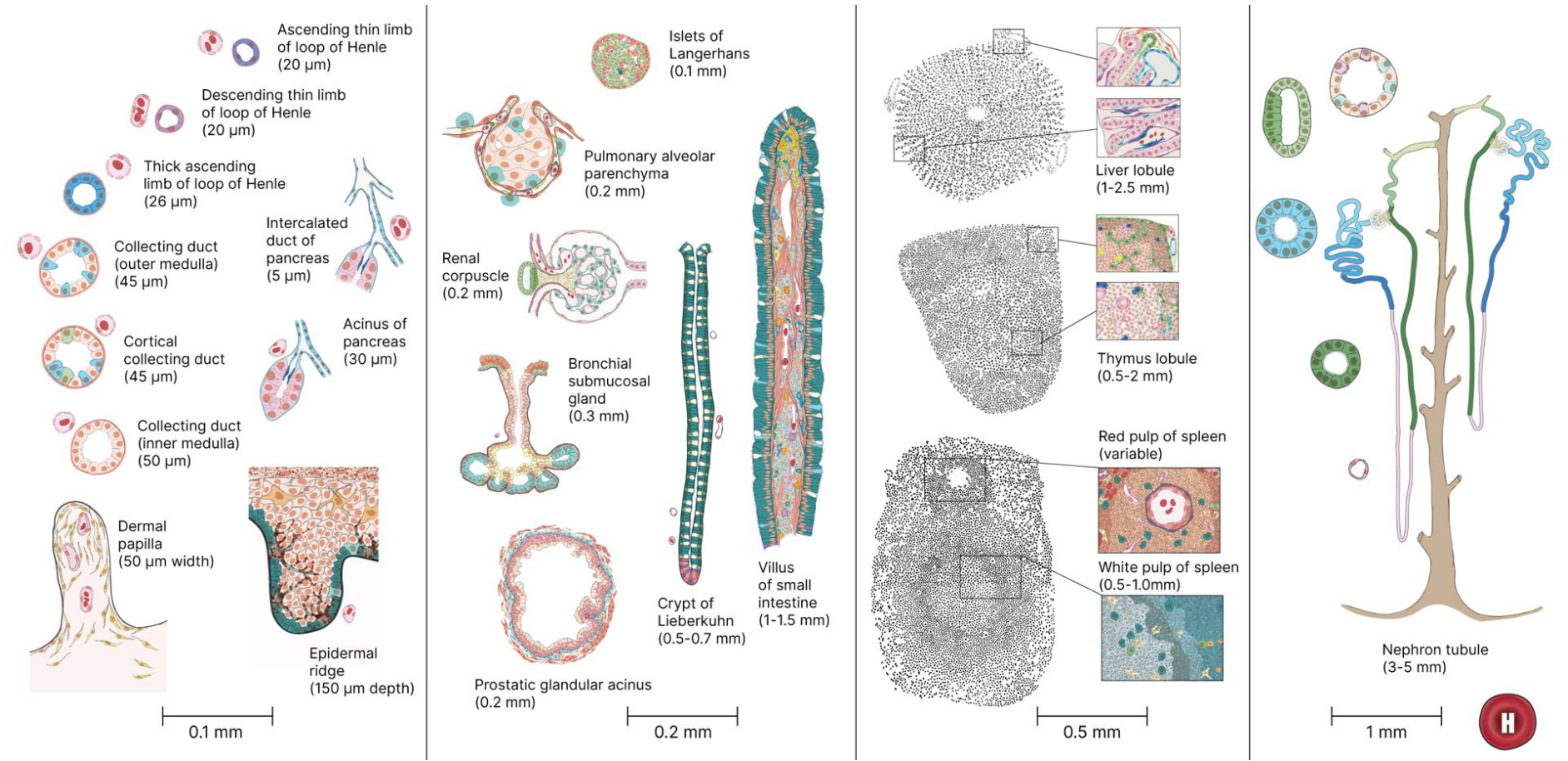
HRApop (Experimental Data + Cell Summaries)

HRAlit (HRA-relevant Literature)

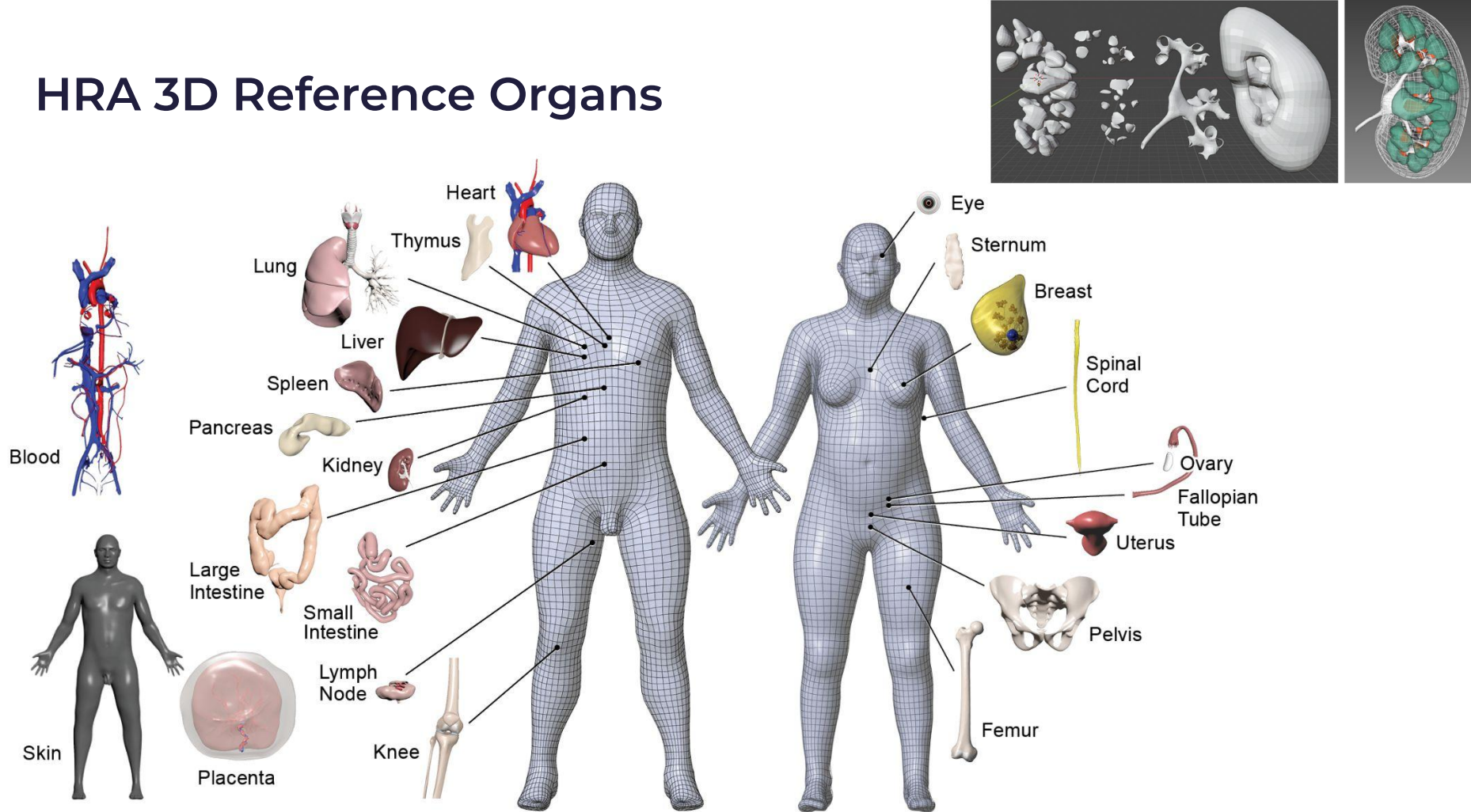
HRA ASCT+B Tables



HRA Functional Tissue Unit (FTU) Illustrations

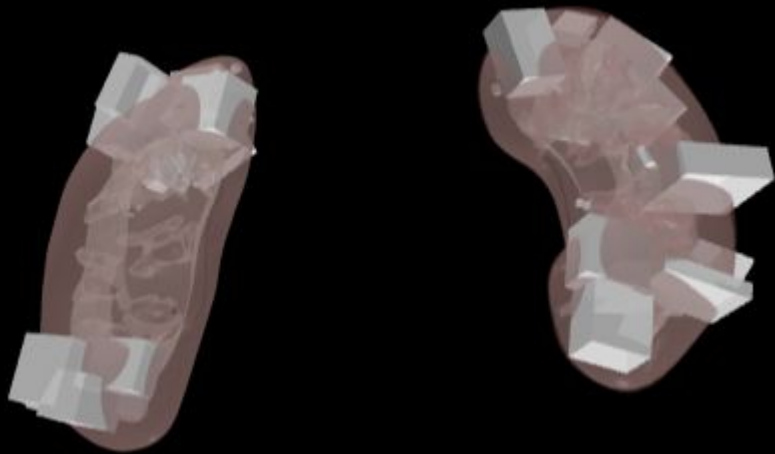


HRA 3D Reference Organs

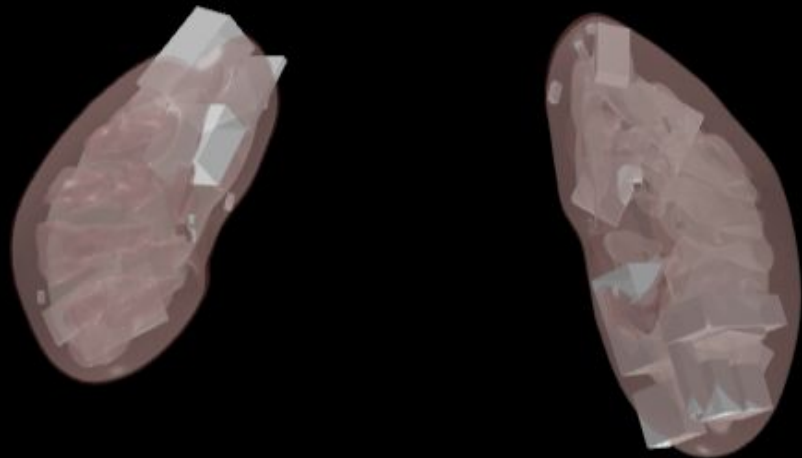


HRA 3D Reference Organs: kidneys

Male

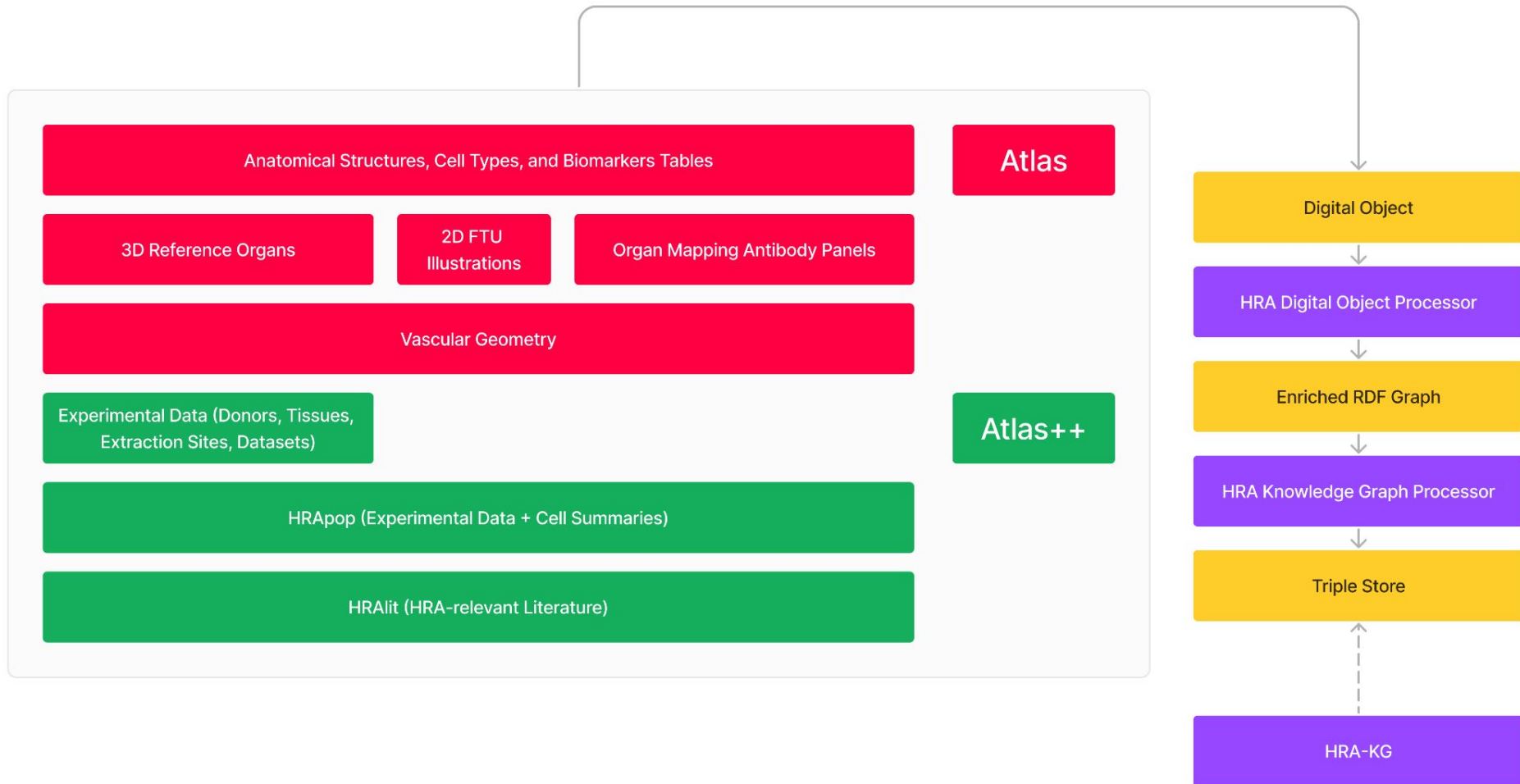


Female

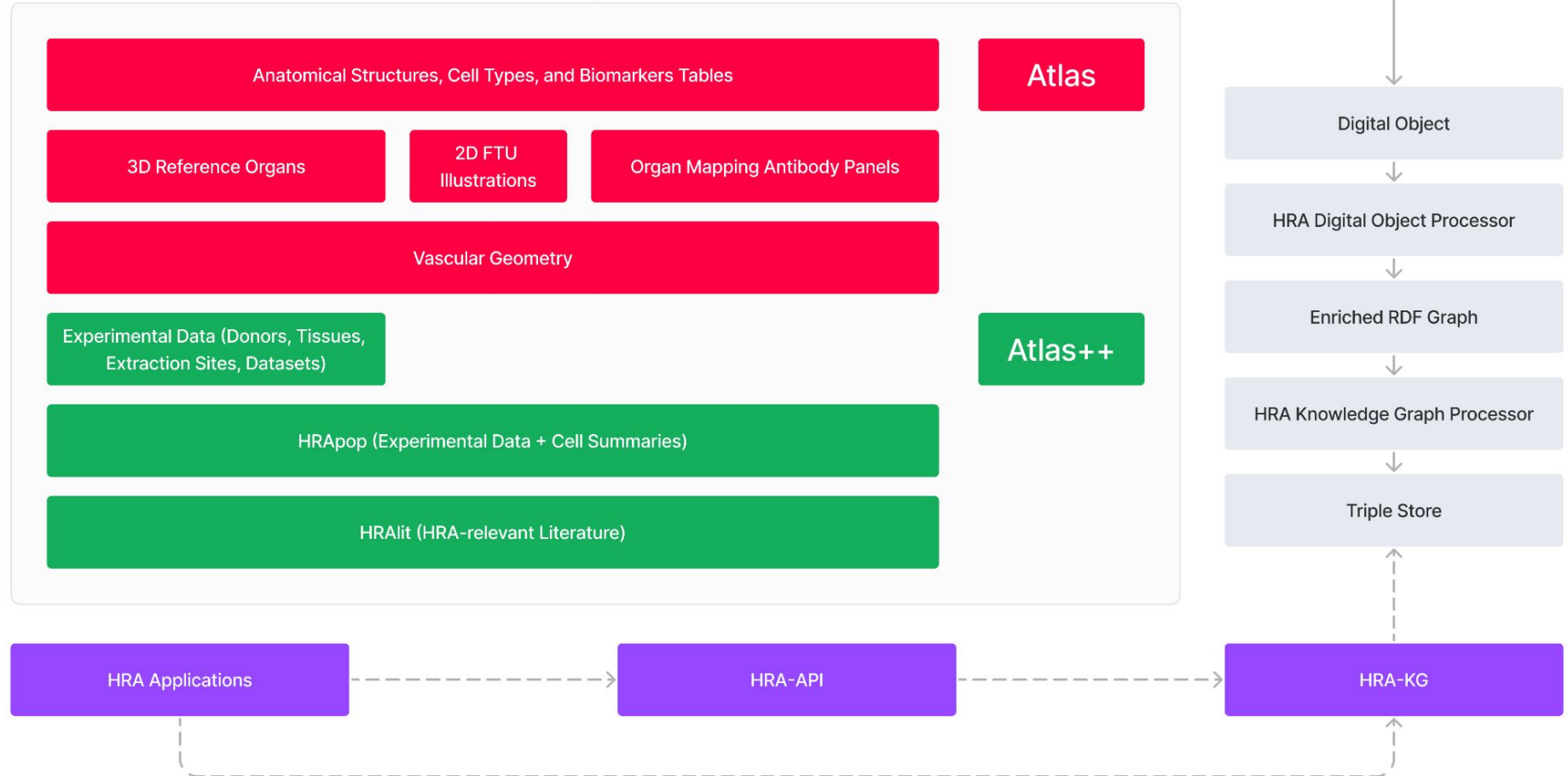


126 tissue blocks with 1,198 datasets from 4 consortia

HRA Knowledge Graph Framework



HRA API and Applications





HRA KG Paper

Title: Construction, Deployment, and Usage of the Human Reference Atlas Knowledge Graph

Link: <https://www.nature.com/articles/s41597-025-05183-6>

Date: July 1, 2025

Topic: This paper presents the **HRA KG v2.2**, which uses 10 ontologies to interlink 33 anatomical structures (ASs), cell types (CTs), plus biomarkers (B) tables, 71 3D reference objects for organs, 22 Functional Tissue Units (FTUs), 11,698 single-cell (sc) datasets, and other **HRA Digital Objects (DOs)**. The HRA KG is accessible via (1) dynamic **queries** through its SPARQL endpoint at lod.humanatlas.io/sparql, (2) the Representational State Transfer (REST)-ful **HRA application programming interface (API)** with **clients** available in JavaScript, TypeScript, Angular 17+ , and Python 3.6+ (humanatlas.io/api), and (3) several **interactive user interfaces (UIs)**. This work presents open code and infrastructure to construct the HRA KG out of disparate data across tabular/non-tabular and nested/flat HRA DOs while ensuring processed data conforms to **5-star Linked Open Data (LOD) principles**. The HRA KG data can also be accessed via content negotiation from lod.humanatlas.io. Any version of any HRA DO can be obtained in the supported RDF format hosted by the HRA KG.

Data Descriptor | [Open access](#) | Published: 01 July 2025

Construction, Deployment, and Usage of the Human Reference Atlas Knowledge Graph

[Andreas Bueckle](#)  [Bruce W. Herr II](#)  [Josef Hardi](#), [Ellen M. Quardokus](#), [Mark A. Musen](#) & [Katy Börner](#) 

[Scientific Data](#) **12**, Article number: 1100 (2025) | [Cite this article](#)

2401 Accesses | **2** Citations | **5** Altmetric | [Metrics](#)

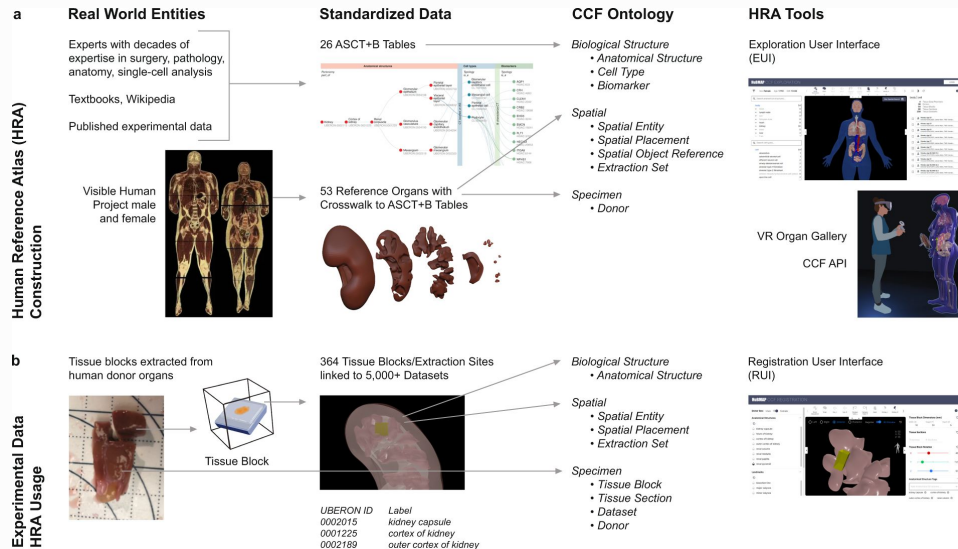
Abstract

The Human Reference Atlas (HRA) for the healthy, adult body is being developed by a team of international, interdisciplinary experts across 25+ consortia. It provides standard terminologies and data structures for describing specimens, biological structures, and spatial positions of experimental datasets and ontology-linked reference anatomical structures (ASs), cell types (CTs), and biomarkers (Bs). This paper introduces the HRA Knowledge Graph (KG) as central data resource for the HRA, supporting cross-scale, biological queries to Resource Description Framework graphs using SPARQL. In May 2025, the HRA KG v2.2 covers 71 organs with 5,800 ASs, 2,268 CTs, 2,531 Bs; it has 10,064,033 nodes, 171,250,177 edges, and a size of 125.84 GB. The HRA KG comprises 13 types of Digital Objects (DOs) using the Common Coordinate Framework Ontology to standardize core concepts and relationships across DOs. This work (1) provides data and code for HRA KG construction; (2) details HRA KG deployment as Linked Open Data; and (3) illustrates HRA KG usage via application programming interfaces, user interfaces, and data products. A companion website is at cns-iu.github.io/hra-kg-supporting-information.

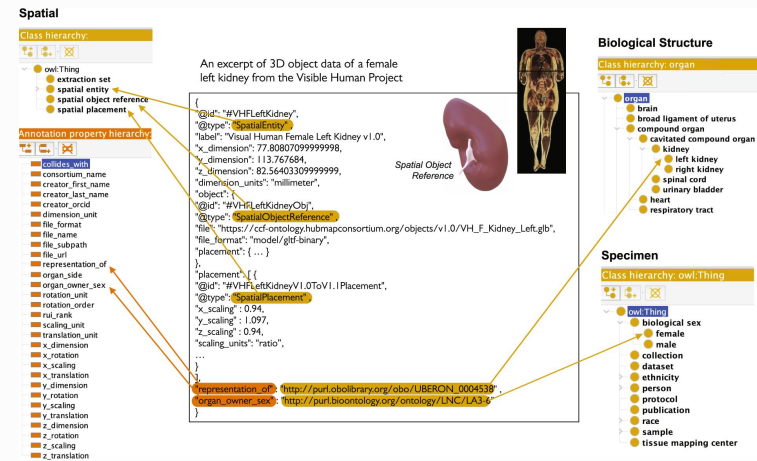
HRA KG and its role in the HRA ecosystem

- **Data backend** for the HRA
- Answer **biological queries across scales** by providing reference information for the healthy, human, adult organs, assay types, cell types, and biomarkers via programmatic, biological queries using SPARQL
- Publishes **13 types of HRA Digital Objects (DOs)** as multiple **Resource Description Framework (RDF)** graphs and uses the **Common Coordinate Framework (CCF) Ontology v3.0**
- Data is provided in raw formats (CSV, GLB, SVG, etc.), processed JSON data, and RDF (JSON-LD, Turtle, RDF/XML, N-Triples, and N-Quads).

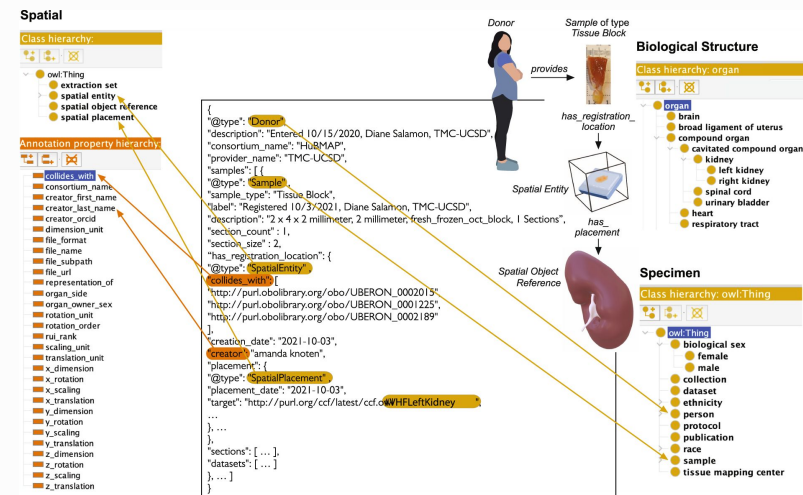
CCF Ontology: Prior Work



HRA KG Construction



HRA KG Usage



Herr, B. W., Hardi, J., Quardokus, E. M., Bueckle, A., Chen, L., Wang, F., Caron, A. R., Osumi-Sutherland, D., Musen, M. A. & Börner, K. Specimen, biological structure, and spatial ontologies in support of a Human Reference Atlas. *Scientific Data* 10, 171 (2023). Link: <https://www.nature.com/articles/s41597-023-01993-8>

HRA KG Data

Digital Objects in HRA v2.4 ?

34

ASCT+B Tables

37

OMAP Tables

26

FTU Illustrations

1,469

3D Anatomical Structure
Models

1

Blood Vessel Segment

35

SOPs

Size of the HRA Knowledge Graph ?

1,855,335

Nodes

189,161,716

Edges

140,187 MB

Size

Ontologies Extended ?

198

Terms added to Uberon

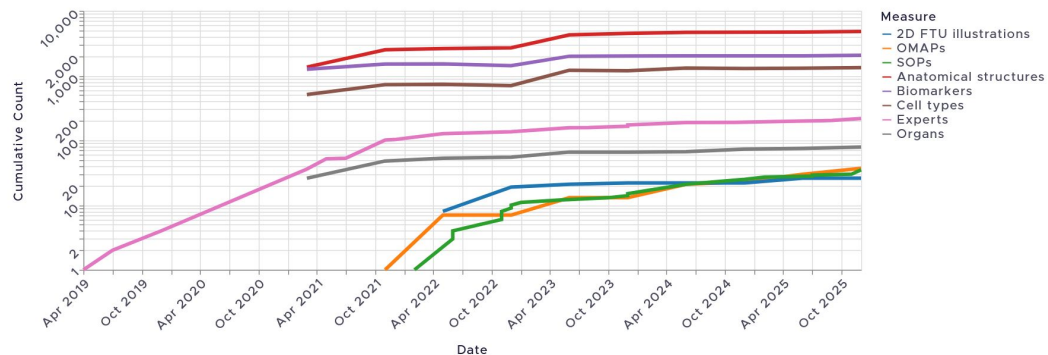
233

Terms added to CL

468

Terms added to PCL

HRA Growth Per Release ?



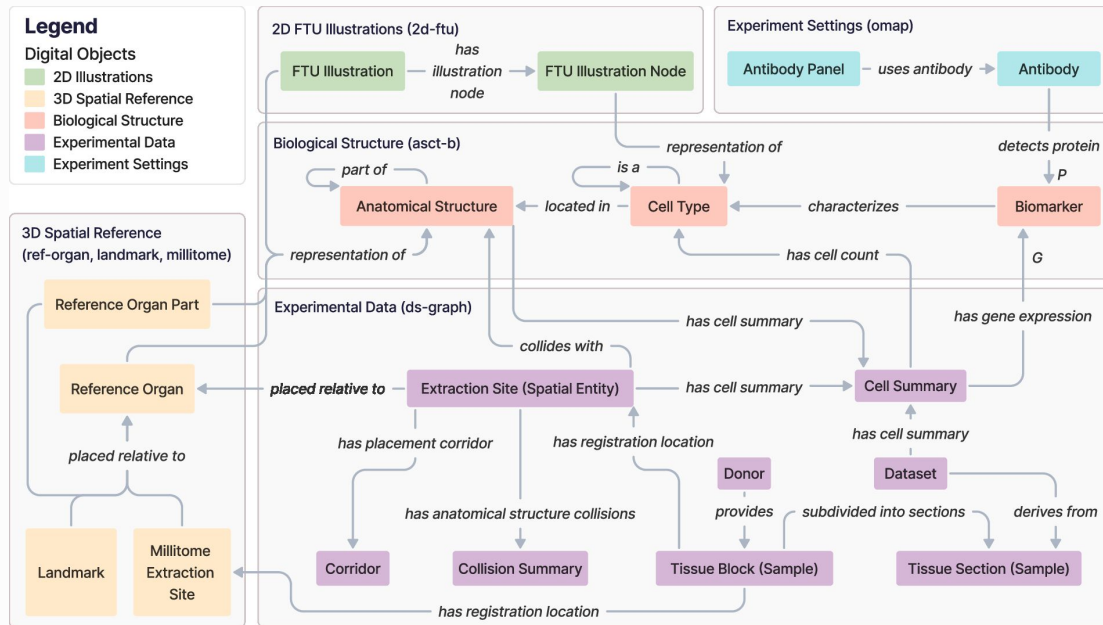
<https://apps.humanatlas.io/dashboard/data>

HRA KG Usage





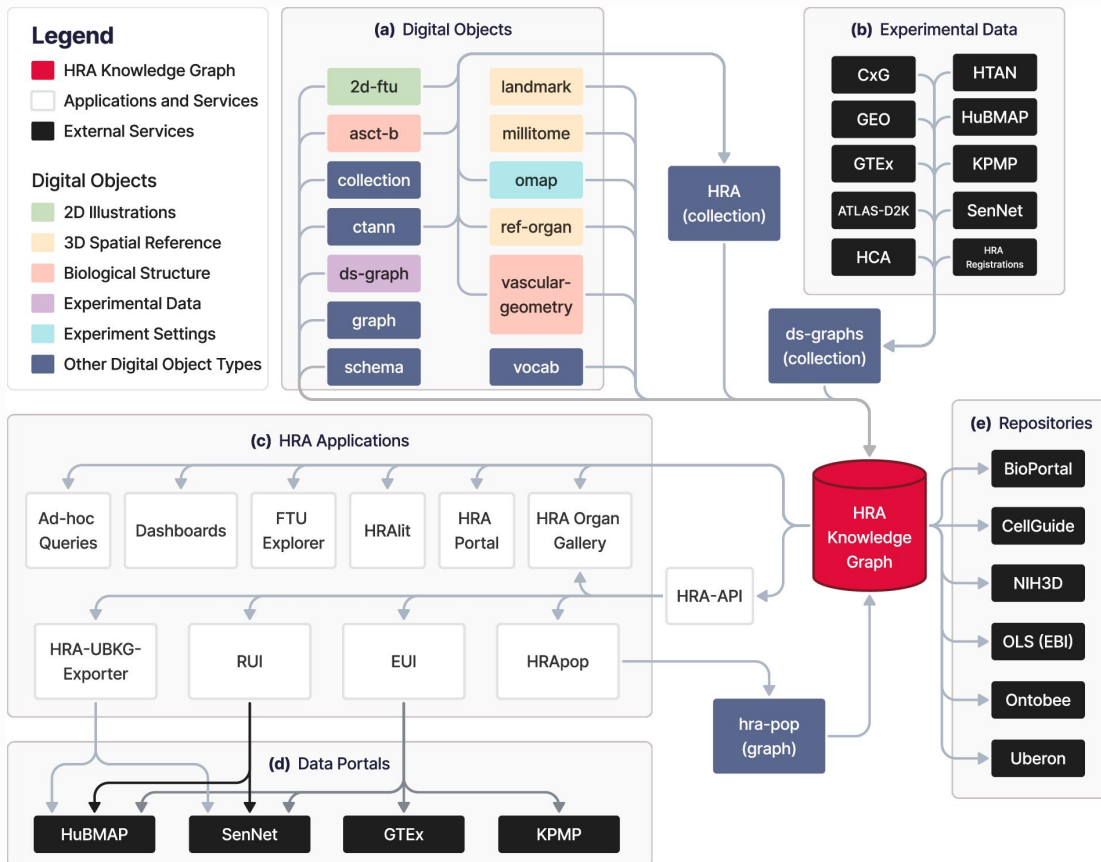
HRA KG Paper



The HRA KG metagraph illustrates high-level relationships between subgraphs. Note that all edge labels have been modified as needed to avoid overlap while maintaining correct semantics. For class names (e.g., FTU Illustration), a blank space was added between PascalCased class names.



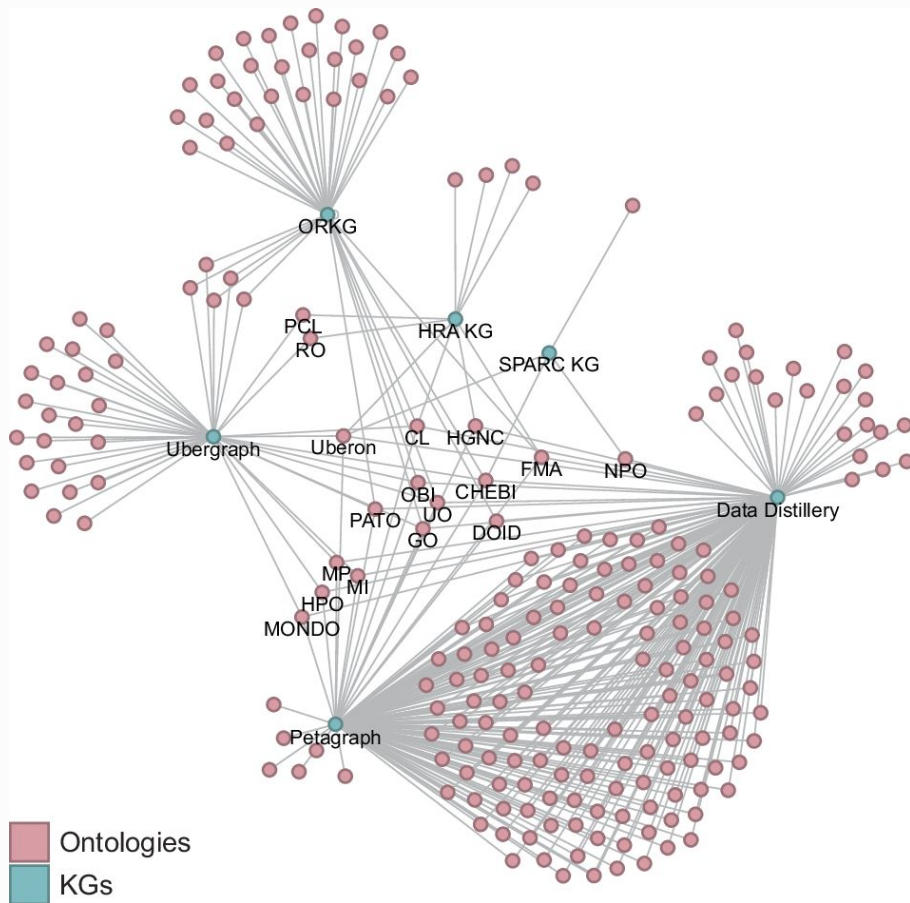
HRA KG Paper



The role of the HRA KG in the HRA data ecosystem: (a) HRA DOs, (b) experimental datasets as well as 3D size, location, and rotation of tissue blocks registered into the HRA, (c) HRA applications, (d) data portals, and (e) external services of the HRA KG. Note that the HRA KG is able to serve all existing versions of the HRA and HRA DOs, including those preceding the most recent HRA v2.2 release in December 2024.



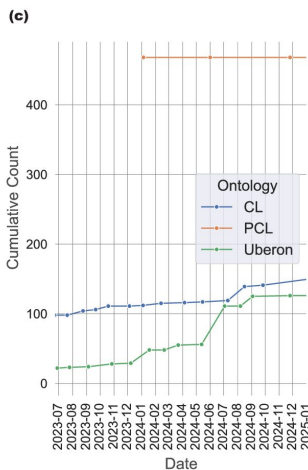
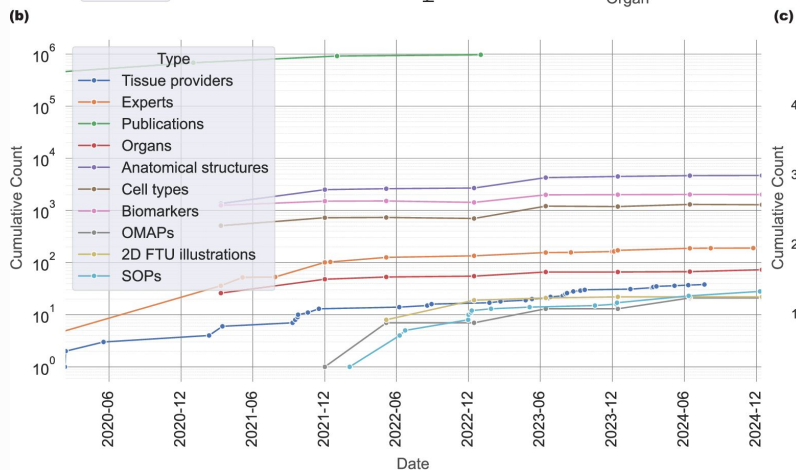
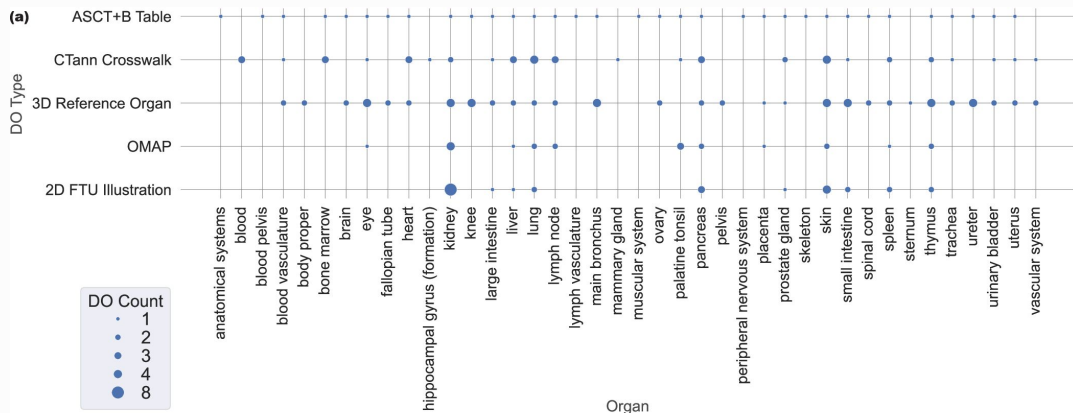
HRA KG Paper



Bimodal network of KGs and the ontologies they ingest and serve. Labels are only shown for nodes with a degree greater than or equal to three, i.e., the KGs themselves and all ontologies shared by at least three KGs. Two ontologies have manually added labels (PCL, RO). Note that Petagraph and Data Distillery ingest many of the same ontologies and are thus very similar. The most shared ontologies across KGs are Uberon, CL, FMA, HGNC, CHEBI, STRING, DOID, UO, PATO, OBI, and GO. The layout was made with Gephi (gephi.org), using the Yifan Hu (Proportional) Algorithm for layout (yifanhu.net/PUB/graph_draw.pdf).



HRA KG Paper



Graphs from the HRA Dashboard showing HRA KG v2.2 growth and coverage: (a) DOs per organ, (b) HRA KG growth in terms of DOs over time, (c) number of terms added from the HRA and other sources to CL, PCL, and Uberon over time.

Spotlight: HRA API SPARQL Queries

- Custom SPARQL queries for the HRA KG
- Deployed via grlc.io, constructs RESTful API from SPARQL queries on GitHub
- Queries live at <https://github.com/hubmapconsortium/ccf-grlc>
- See <https://apps.humanatlas.io/api/grlc>

HRA-API SPARQL queries (hra)

HuBMAP Help Desk help@hubmapconsortium.org | License: [MIT License](#)

This API provides custom SPARQL queries using the Human Reference Atlas (HRA) Knowledge Graph (HRA-KG). See the [HuBMAP HRA Portal](#) for details.

Queries powered by [grlc.io](#), the git repository linked data API constructor.

API SERVER

- ☒ <https://apps.humanatlas.io/api/grlc/hra> - HRA-API Production
- ☐ <https://apps.humanatlas.io/api--staging/api/grlc/hra> - HRA-API Staging
- ☐ <https://apps.humanatlas.io/grlc/hra> - Local Server

SELECTED: <https://apps.humanatlas.io/api/grlc/hra>

[Expand all](#) | [Collapse all](#) sections

General 8

GET	/2d-ftu-parts	get 2D FTU images and parts
GET	/2d-ftu-parts-unique	get 2D FTU images and parts that are unique to the ftu
GET	/2d-ftu-parts-unique-hra	get 2D FTU images and parts that are unique to the ftu (via ASCT+B relationships)
GET	/2d-ftu-parts-unique-hra-obo	get 2D FTU images and parts that are unique to the ftu (via ASCT+B and OBO relationships)
GET	/all-digital-objects	Every digital object pushed to lod.humanatlas.io

Spotlight: HRA KG Explorer

- A new interface for exploring published HRA Digital Objects
- Filtering, faceted search, and detailed DO metadata pages.
- Go to <https://lod.humanatlas.io> to get started

The screenshot shows the 'Apps / Knowledge Graph' page of the Human Reference Atlas. On the left, a 'Filters' sidebar lists categories: Digital objects, HRA release version, Organs, Anatomical structures, Cell types, and Biomarkers. A search bar is positioned above a table of results. A dropdown menu is open from the search bar, showing a list of search results with counts: 3D Organs (80), ASCT+B Tables (41), Cell Type Annotations (9), Collections (5), Dataset Graphs (7), and Functional Tissue Unit Illustrati... (27). The main table displays columns for Type, Organ, #Cell types, #Biomarker types, and Date last published. It lists several entries, including 'Common Coordinate Framework', '2D Bronchial Submucosal Gland FTU for Lung', and 'Top of Henle FTU for Kidney'. The top right of the page shows navigation links for Data, Apps, Training, and About, along with a user icon.

The screenshot shows the detailed metadata page for a '3D Reference Organ Set for United, Male'. The page header includes the Human Reference Atlas logo and navigation links. Below the title, there is a URL 'https://purl.humanatlas.io/ref-organ/united-male' and a 'Copy' button. A 3D visualization of a human torso is displayed. On the right, a 'Digital object version' dropdown is set to 'v1.0', with a 'Download options' button below it. A table at the bottom provides provenance and metadata details.

Provenance	Metadata
Creator(s)	• Kristen Browne • Heidi Schiehllein
Project lead(s)	Katy Börner
Reviewer(s)	• Bruce W. Herr II • Ellen M. Quardokus
DOI	https://doi.org/10.48539/HBM447.LMRP.747

HRA Cell Type Populations (HRApop)





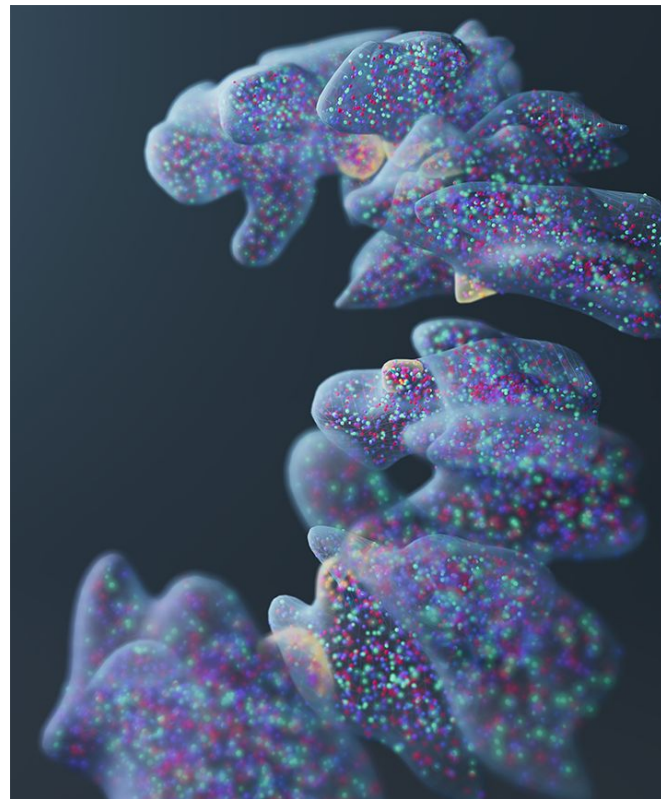
What is a “cell type population”?

Cell type (CT) population:

- Unique CTs and their counts plus mean biomarker expression values computed for anatomical structures (ASs), extraction sites, and datasets.
- Computed from CT counts in high-quality experimental datasets, obtained via cell type annotation tools (Azimuth, CellTypist, popV).
- CT populations are associated with exactly one cell type annotation tool; multiple tools lead to different CT populations.

Source:

<https://www.nature.com/articles/s41597-026-06642-4>





HRApop

scientific **data**

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Data Descriptor | [Open access](#) | Published: 19 March 2026

Cell Type Populations for 3D Anatomical Structures of the Human Reference Atlas

[Andreas Bueckle](#) , [Bruce W. Herr II](#) , [Lu Chen](#), [Daniel Bolin](#), [Danial Qaurooni](#), [Michael Ginda](#),
[Yashvardhan Jain](#), [Aleix Puig-Barbe](#), [Kristin Ardlie](#), [Fusheng Wang](#) & [Katy Börner](#) 

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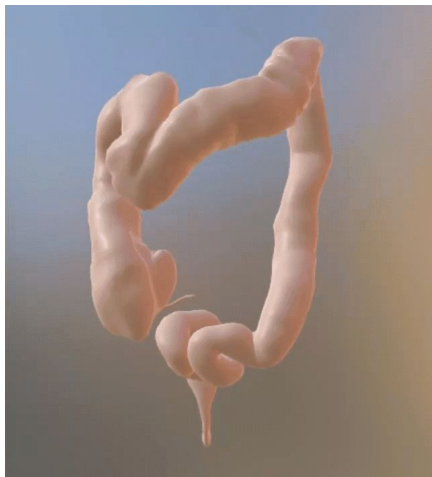
We are providing an unedited version of this manuscript to give early access to its findings.
Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.



<https://www.nature.com/articles/s41597-026-06642-4>

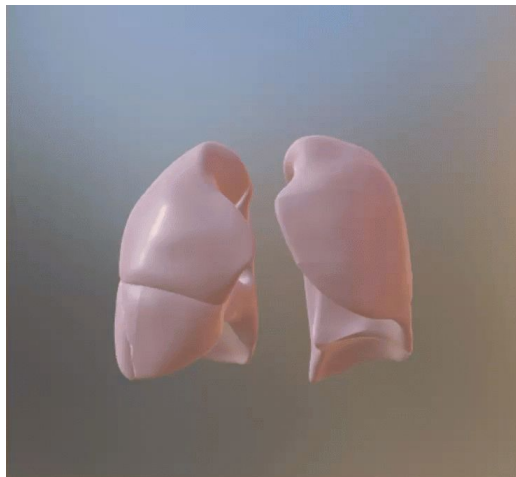


3D Reference Organs



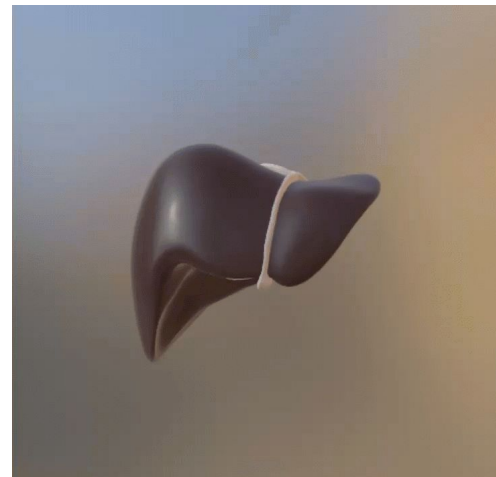
Large intestine

Sex: male
11 ASs



Lung

Sex: female
70 ASs



Liver

Sex: male
23 ASs



Scene Explorer

Filter



Scene

Nodes

__root__

VH_M_colon

VH_M_ascending_colon

VH_M_descending_colon

VH_M_hepatic_flexure_of_colon

VH_M_rectum

VH_M_sigmoid_colon

VH_M_splenic_flexure_of_colon

VH_M_transverse_colon

VH_M_ileocecal_valve

VH_M_caecum

VH_M_vermiform_appendix

default camera

hdrSkyBox

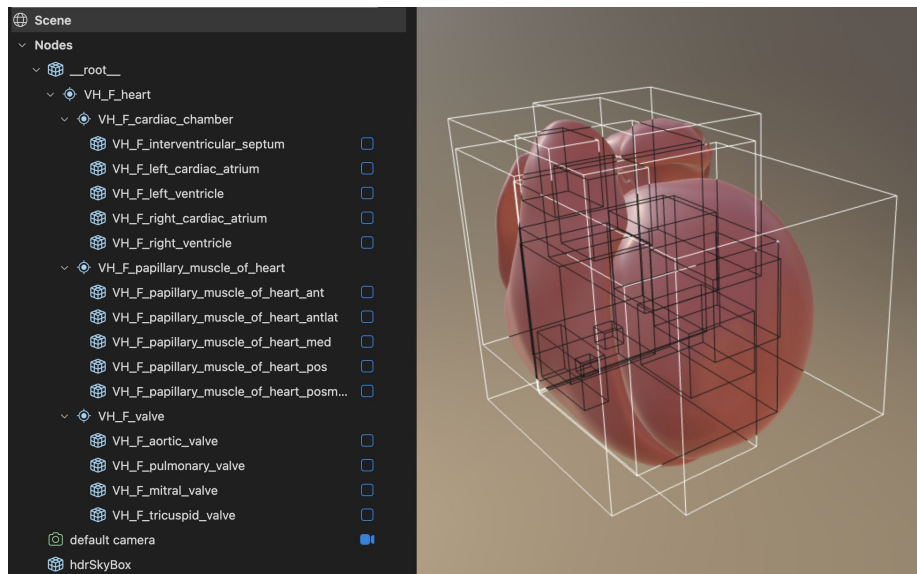
> Materials

> Textures





Tie to Spatial Data: 3D Reference Objects



Custom built by our medical illustrator team
with input from subject matter experts

Support HRA applications, incl. HRA Organ
Gallery

ASs are labeled with ontology IDs

Explore all at

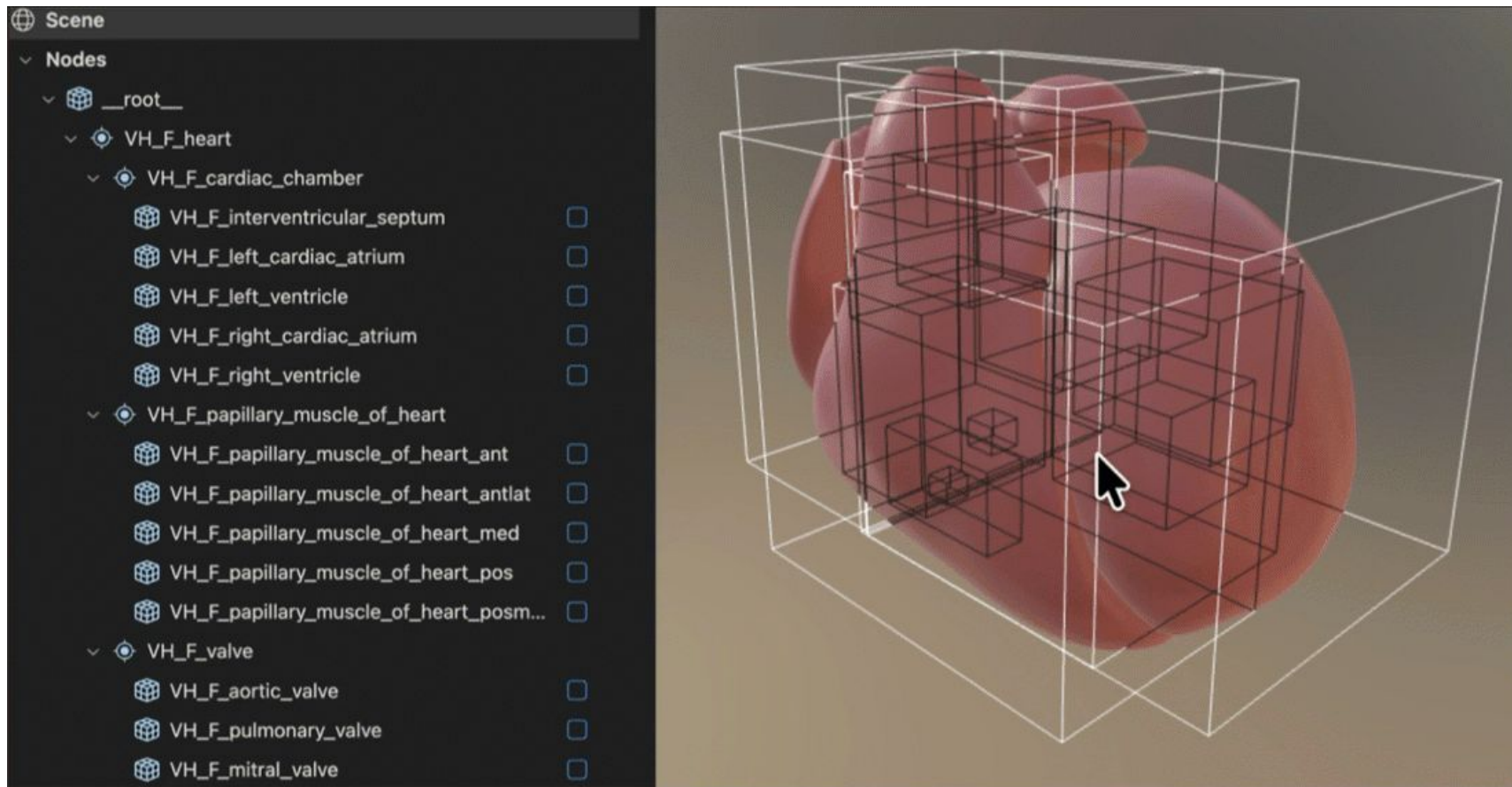
<https://humanatlas.io/3d-reference-library>

and

<https://3d.nih.gov/collections/hra>



3D Reference Organs





Registration User Interface (RUI)

Registration User Interface

Opacity Settings ⋮

Anatomical Structures

- All anatomical structures
- Interventricular septum
- Left cardiac atrium
- Heart left ventricle
- Right cardiac atrium
- Heart right ventricle
- Papillary muscle of heart
- Anterior papillary muscle of left ventricle
- Anterolateral head of lateral papillary muscle of le...
- Septal papillary muscle of right ventricle
- Posterior papillary muscle of right ventricle
- Posteromedial head of posterior papillary muscle ...
- Valve
- Aortic valve
- Pulmonary valve
- Mitral valve
- Tricuspid valve

Landmarks

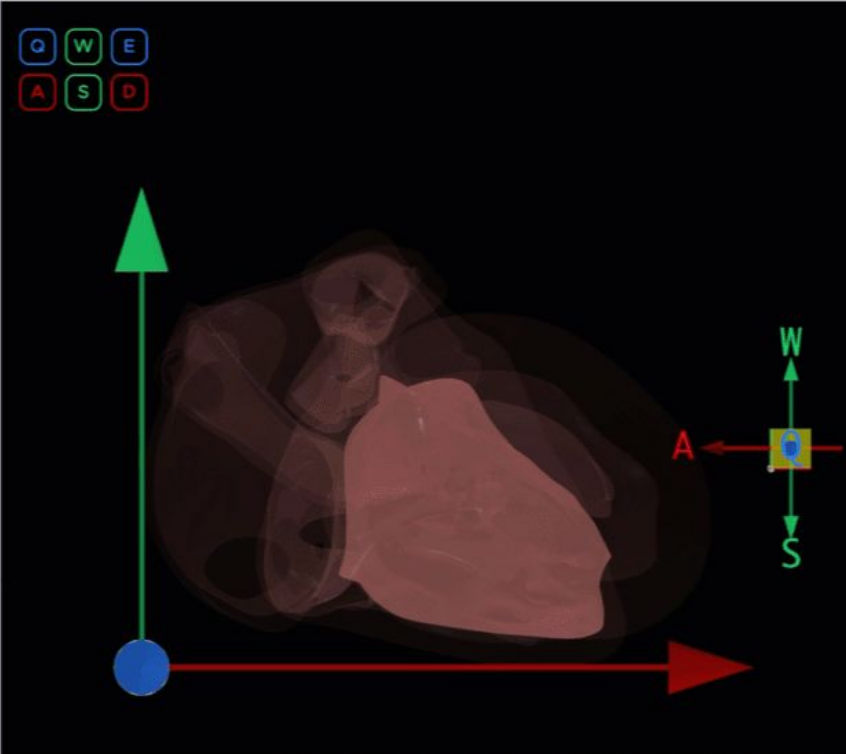
- All landmarks
- Heart arteries
- Heart veins

Registration ⋮

Left Right Anterior Posterior Register Preview

Q W E

A S D



Tissue Block Settings ⋮ ?

Position Millimeters	Size Millimeters	Rotation Degrees
X 162	Width 10	x 0
Y 54	Height 10	y 0
Z 50	Depth 10	z 0

Tissue Section

Thickness (μm) 0	#Sections 0
---------------------	----------------

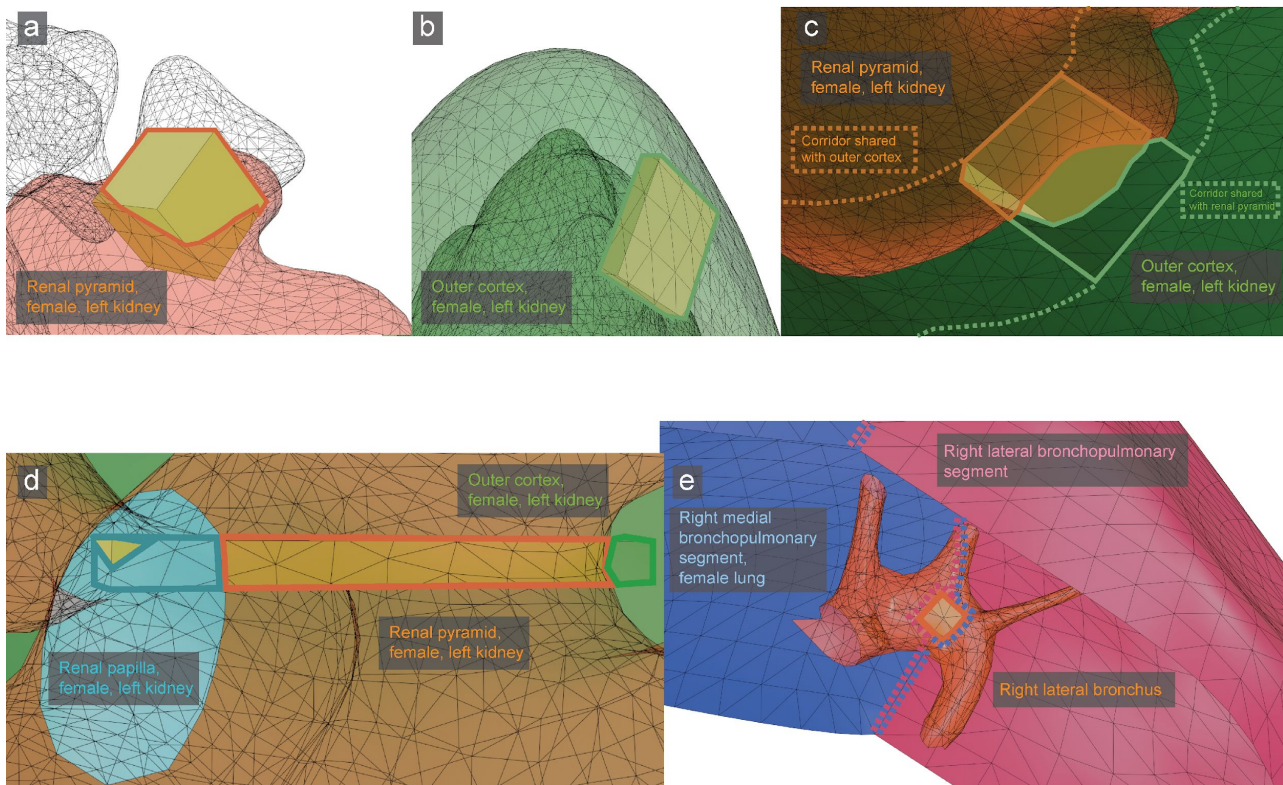
Anatomical Structure Tags ⋮

Edit Metadata Review and Download

<https://apps.humanatlas.io/rui/>



Intersections between tissue blocks and ASs





Computing HRApop

Legend

Computation

- Applications and Services
- Data Portals + Other Sources
- Cell Type Annotation Tools
- Extraction Site 1
- Extraction Site 2
- Cell Type 1
- Cell Type 2
- Cell Type 3
- Cell Type 4

Usage

- Donor Metadata
- HRApop Atlas Data

(a) Data Portals and HRA Registrations

CELLxGENE

GTEX

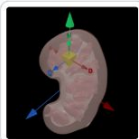
HuBMAP

SenNet

HRA Registrations

(d) RUI Registration

RUI



(b) Donor

Age + Health

BMI

Race/Ethnicity

Sex

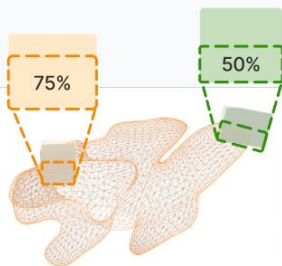
(c) CTann

Azimuth

CellTypist

popV

(e) Extraction Site



(f) Cell Type Population for Dataset

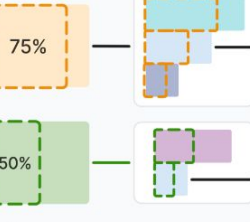
Dataset 1



Dataset 2



(g) % of Tissue Block intersecting with AS



(h) ASpop



(i) HRApop Atlas Data



(j) HRA User Interfaces and Applications

Ad-Hoc Queries

Any non-HRA application

HRA API

HRA Organ Gallery

HRApop Visualizer

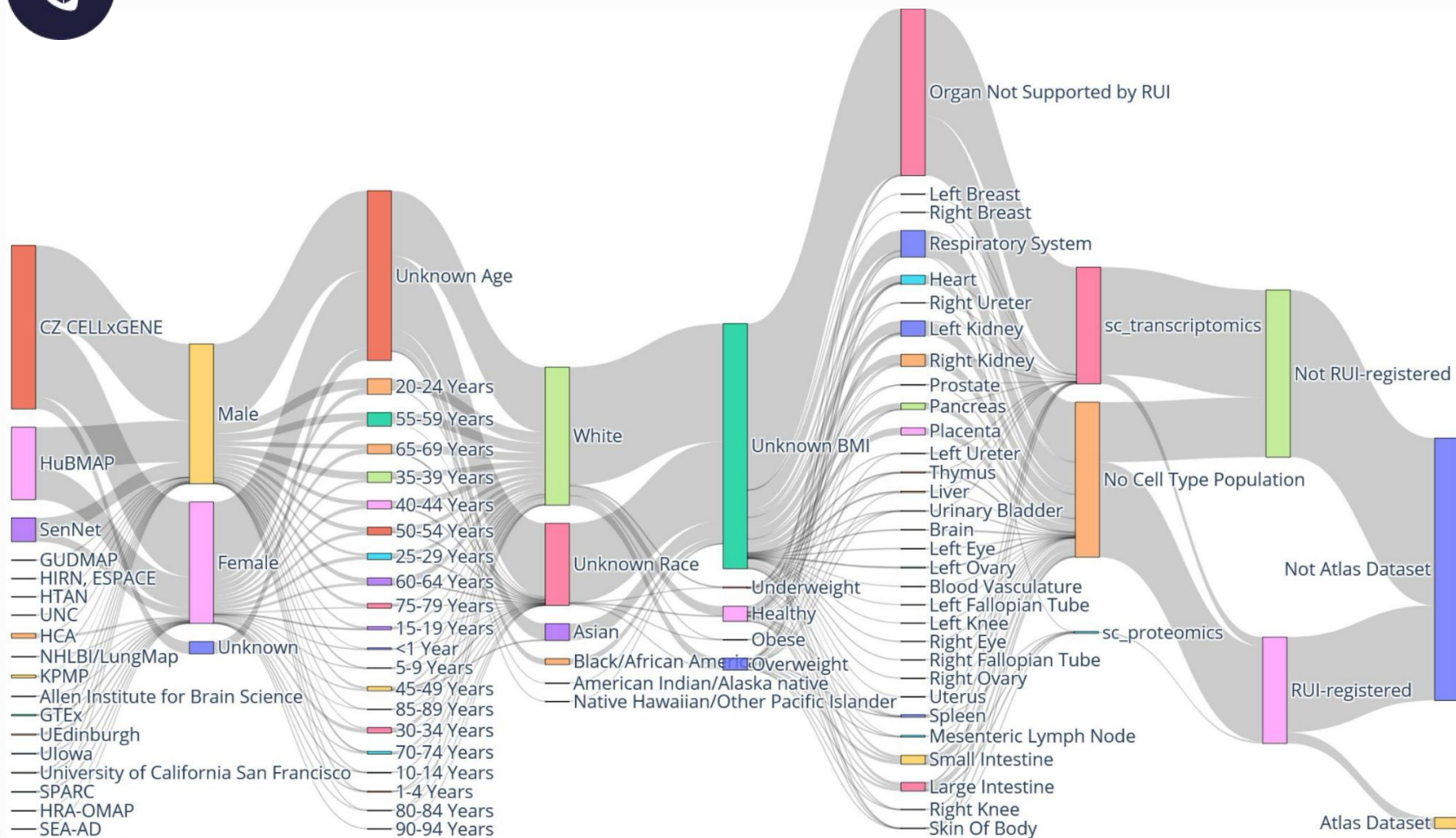
FTU Explorer

Download and Cell Type Annotation (DCTA) Workflow

RUI to Compile Cell Type Populations (RUI2CTpop) Workflow

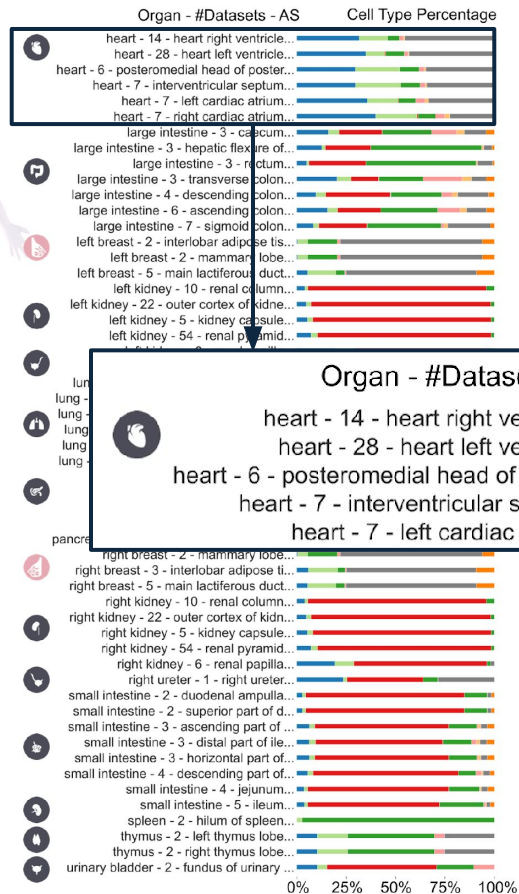
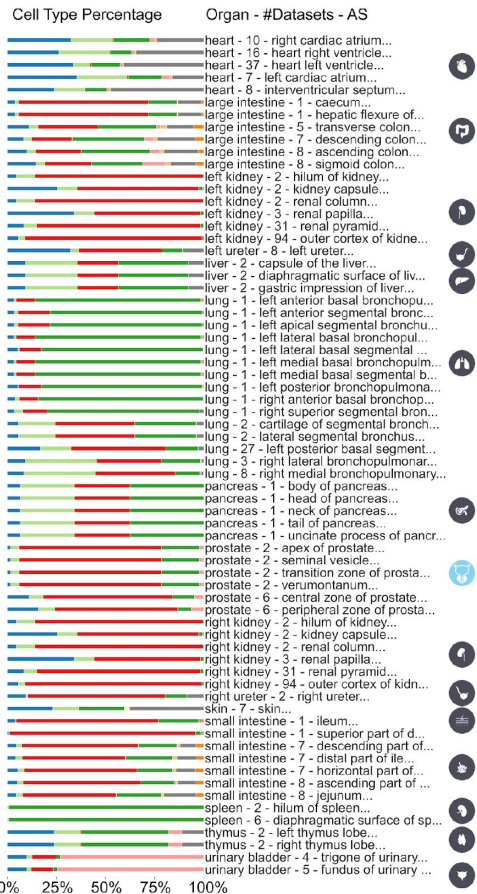


HRApop Atlas Data

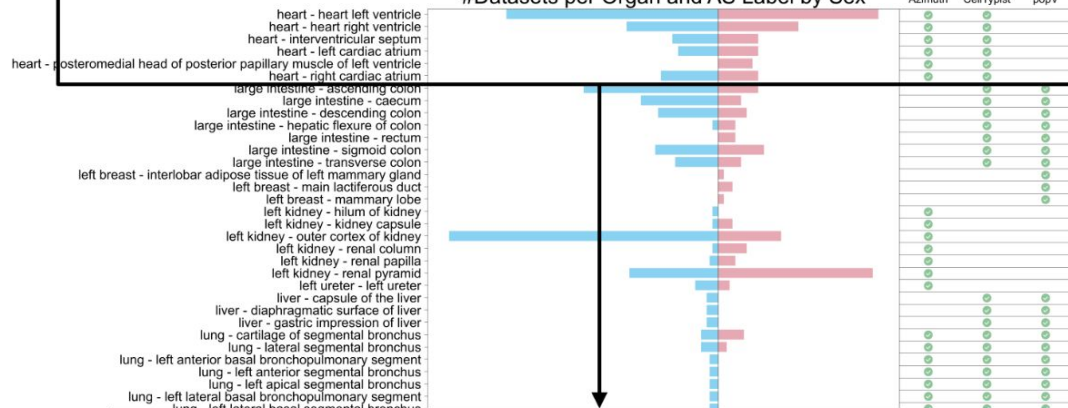




On the AS Level



#Datasets per Organ and AS Label by Sex



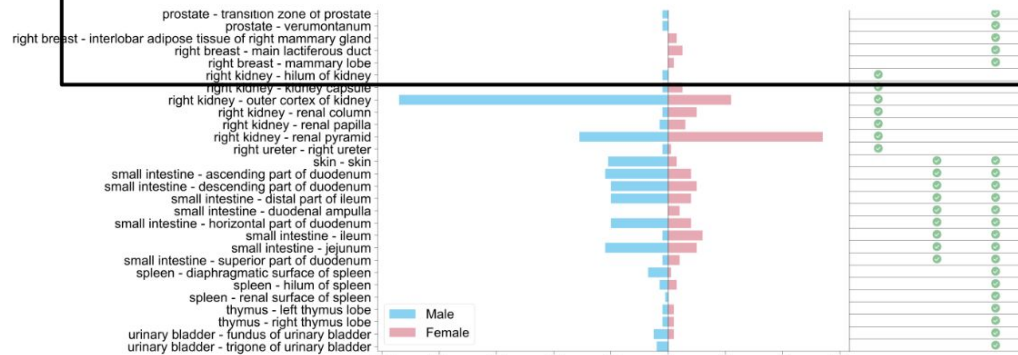
#Datasets per Organ and AS Label by Sex

Azimuth CellTypist popV

heart - heart left ventricle
heart - heart right ventricle
heart - interventricular septum
heart - left cardiac atrium
heart - posteromedial head of posterior papillary muscle of left ventricle
heart - right cardiac atrium

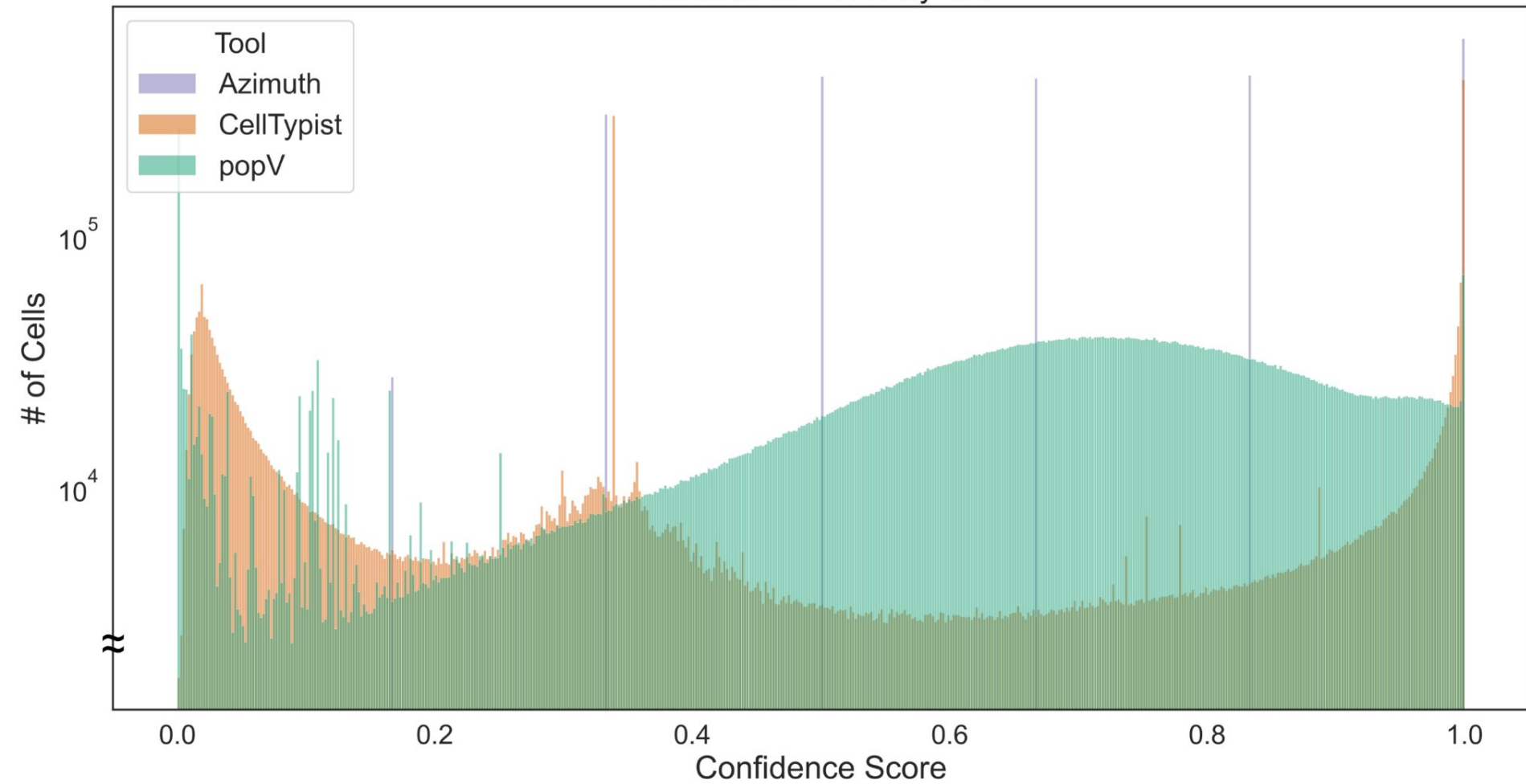


Number of Datasets

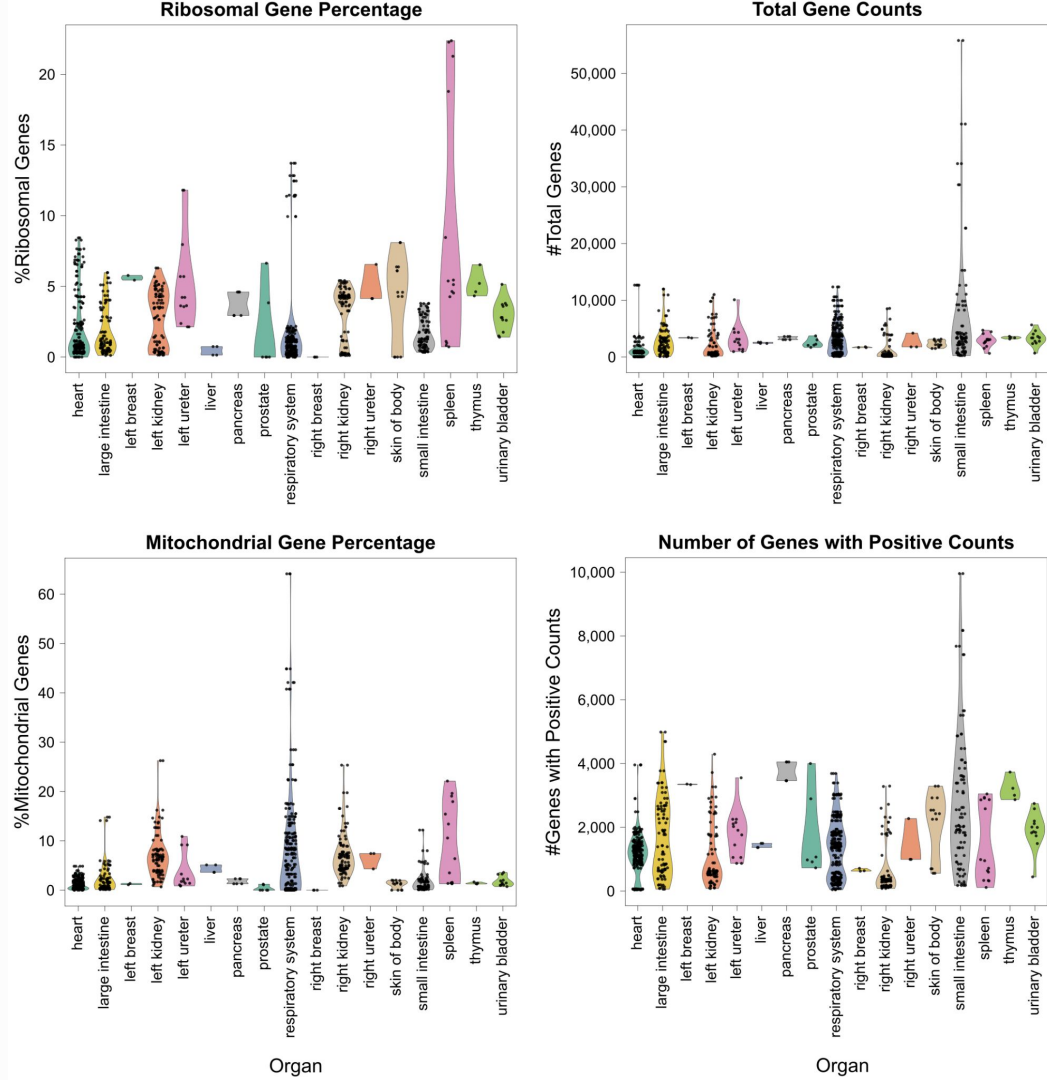


Number of Datasets

Confidence Score by Tool

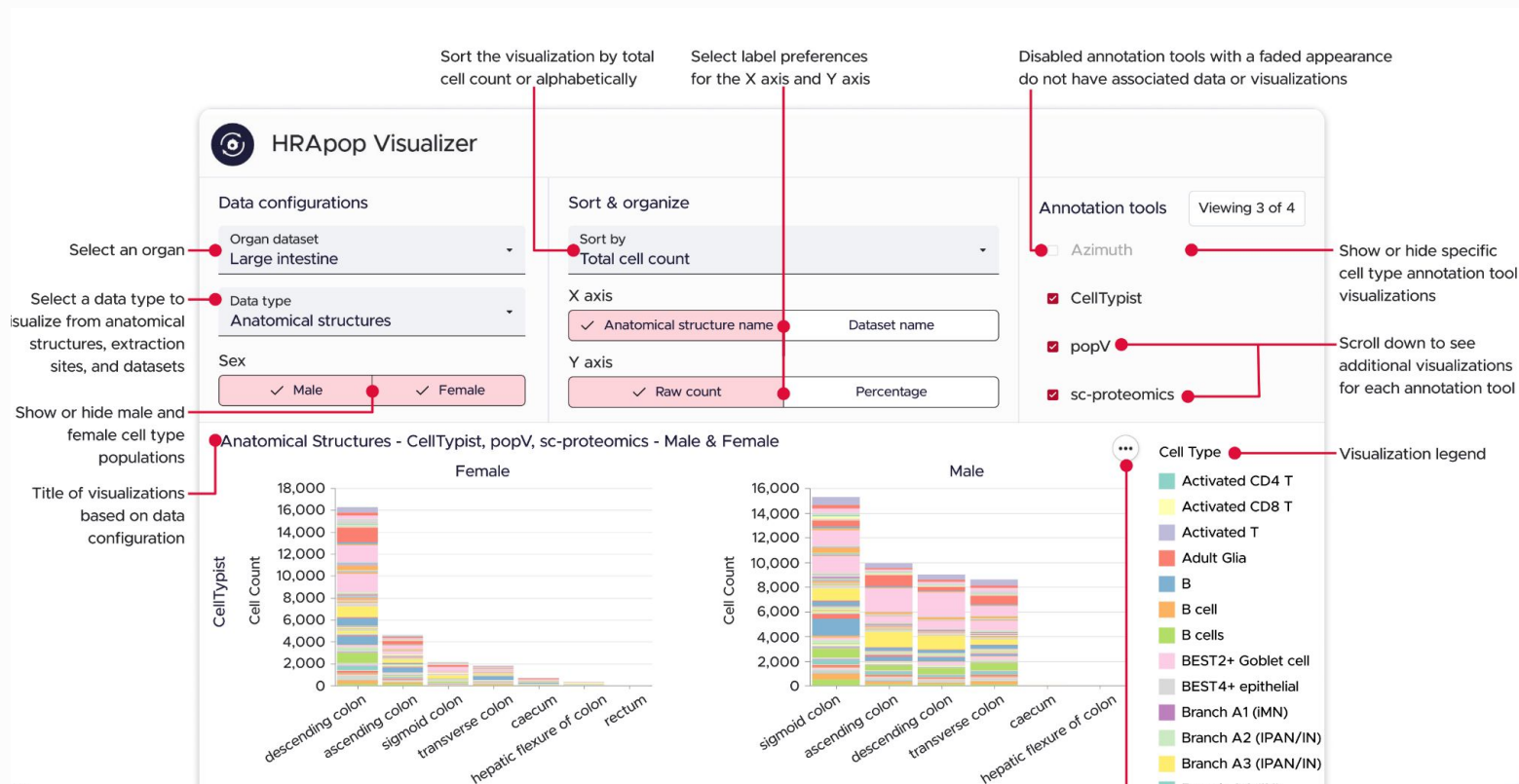


Violin plots with jittered dots for the ribosomal and mitochondrial gene percentage, plus genes with positive counts and total gene counts per organ.





HRApop Visualizer



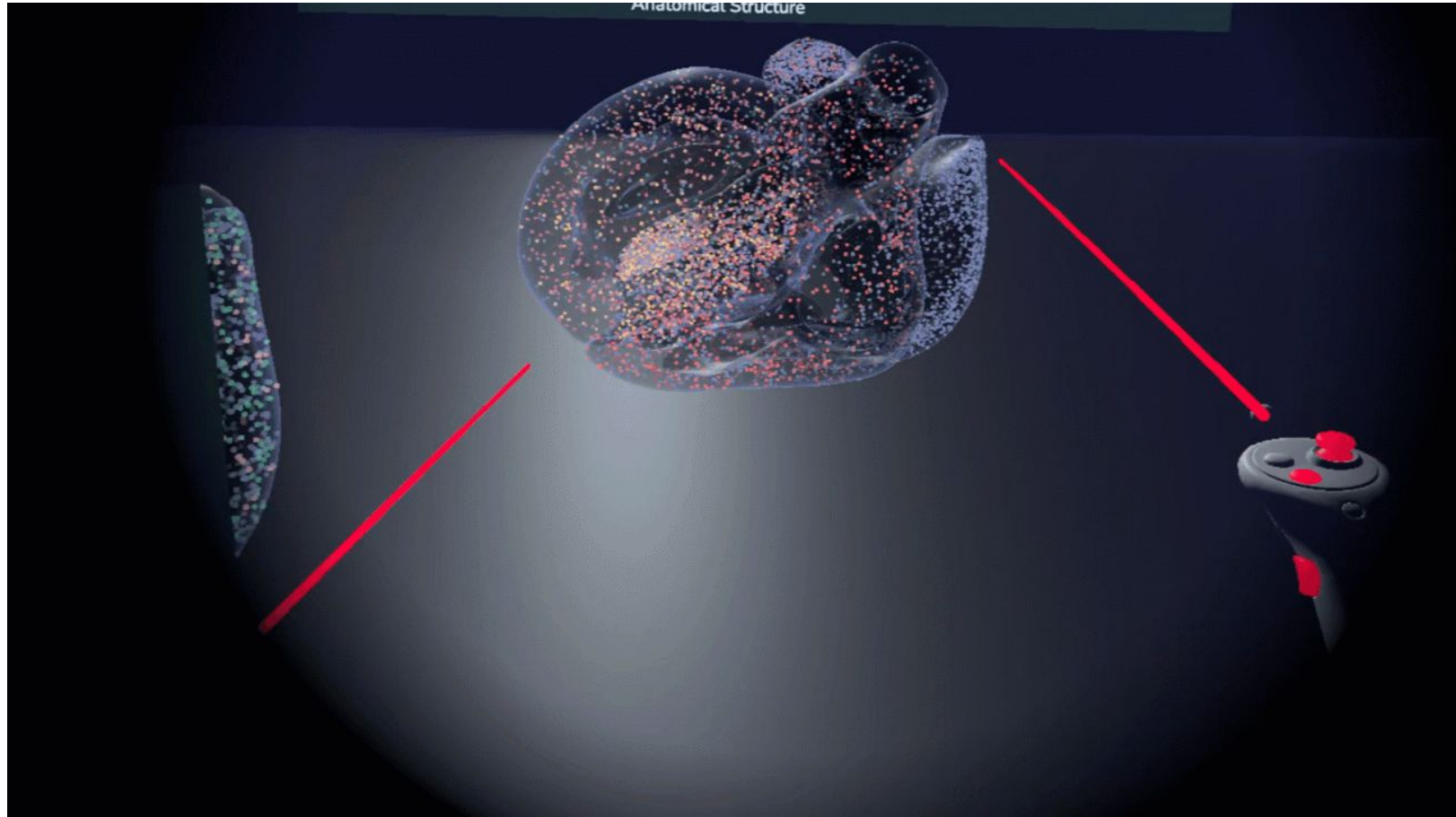


Datasets from HuBMAP, SenNet, GTEx in HRApop v1.0

Sex	Consortium	#Datasets	#Cells	Modality
Female	GTEx	7	47,863	sc_transcriptomics
Male	GTEx	8	70,113	sc_transcriptomics
Female	Human Cell Atlas	63	364,993	sc_transcriptomics
Male	Human Cell Atlas	58	359,273	sc_transcriptomics
Female	HuBMAP	22	900,547	sc_proteomics
Female	HuBMAP	112	2,933,015	sc_transcriptomics
Male	HuBMAP	82	15,676,316	sc_proteomics
Male	HuBMAP	253	6,626,620	sc_transcriptomics
Female	NHLBI/LungMap	1	4,680	sc_transcriptomics
Male	NHLBI/LungMap	2	5,713	sc_transcriptomics
Female	SenNet	18	225,940	sc_transcriptomics
Male	SenNet	36	404,540	sc_transcriptomics
TOTAL		662	27,619,613 sc-transcriptomics: 11,042,750 sc-proteomics: 16,576,863	

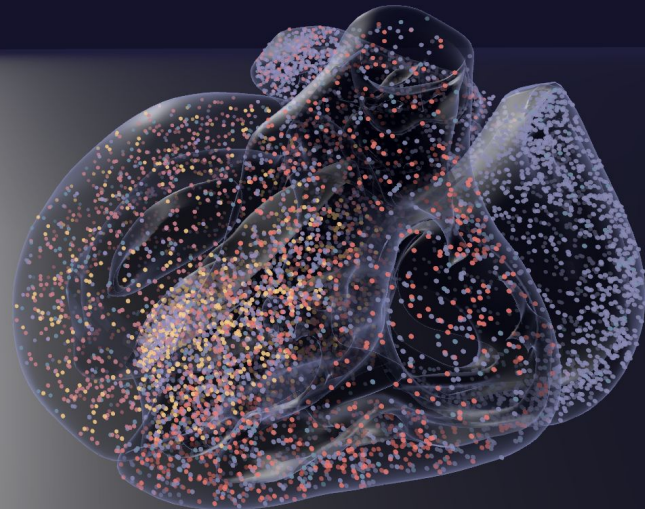
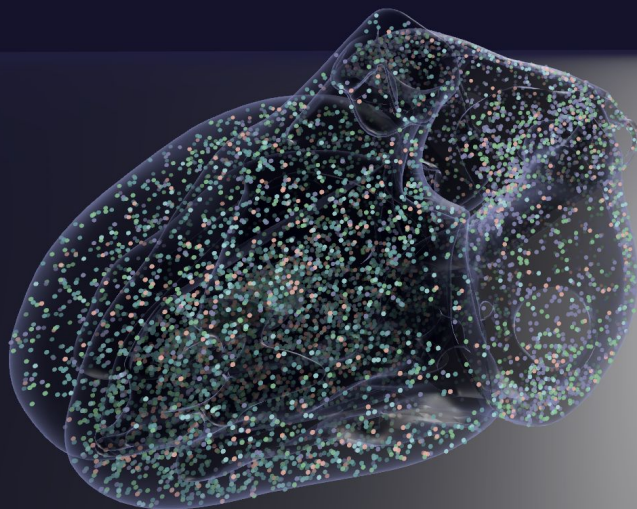
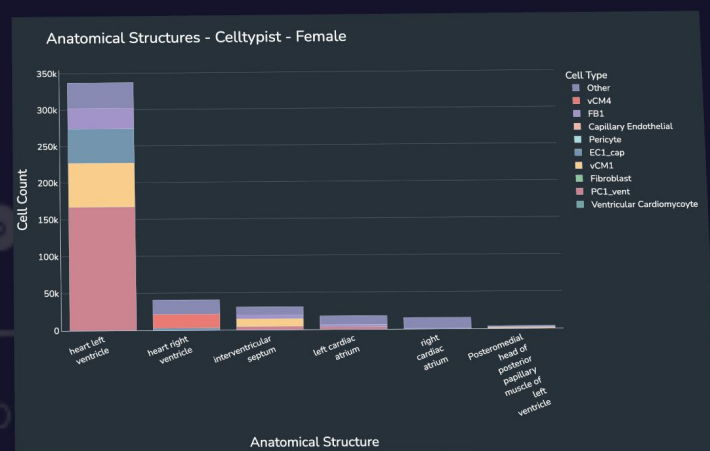
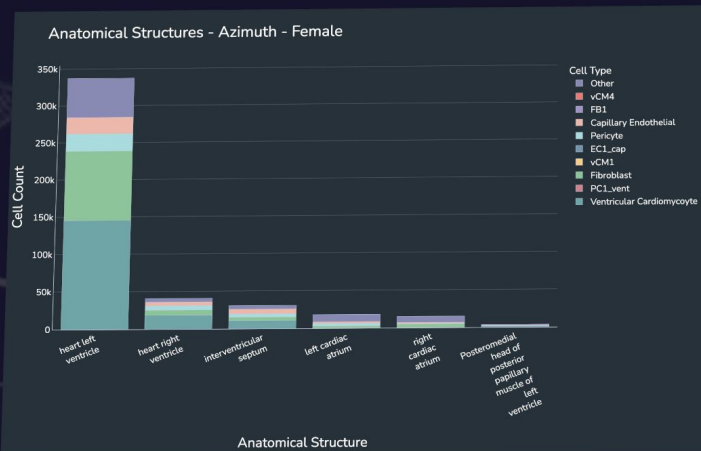


Explore Cell Type Populations by Tool in Virtual Reality



Cell Type

- Other
- vCM4
- FB1
- Capillary Endothelial
- Pericyte
- EC1_cap
- vCM1
- Fibroblast
- PC1_vent
- Ventricular Cardiomyocyte





Next Steps for HRApop

- HRApop paper to be published soon in *Nature Scientific Data*
- Preparing HRApop v1.1:
 - 3x as much data, including DISCO, 47M spatial omics cells
 - More tools: [FR-Match](#), [Pan-Human Azimuth](#), gene expressions with [NS-Forest](#)
 - Data will be fully deployed to **HRA Organ Gallery** once HRApop v1.1 is published.
 - Data usage in
 - HRA FTU Explorer (compare healthy and diseased, age, sex)
 - HPO collab (add disease into HRA/CCF ontology)
 - WPP (physiological function in addition to HRA structure)



Andi's VR Team

Research Assistants (L to R):

Juhi Khare

Yash Kumar

Keyur Parekh (former)

Paeon Luby

Siddhi Chavan

Medical Illustrator (bottom):

Ushma Patel



MC-IU Team

Connecting people is key to our success. Here are some of our great collaborators (apologies to those we missed!)

Principal Investigator, Co-Principal Investigators, and Consultants



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MC-IU PI
CNS Director



Mark Musen
Professor of Medicine
(Biomedical Informatics) and
of Biomedical Data Science



Helen Parkinson
Ontologist



David Van Valen
Assistant Professor of
Biology and Biological
Engineering & Investigator



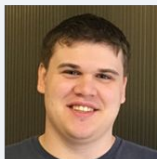
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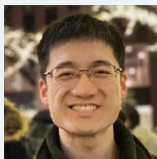
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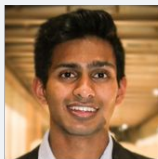
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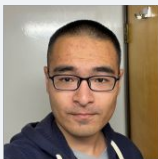
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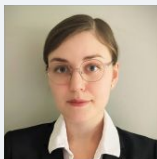
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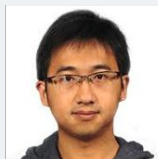
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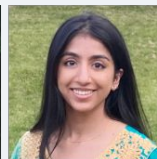
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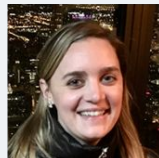
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Human Reference Atlas

Collaborators

HuBMAP



SenNet



GTEx



KPMP



GUDMAP



WHOLE
PERSON
PHYSIOME

+19 Other Consortia

250+ subject matter experts

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More information about the Human Reference Atlas is available at:

<https://humanatlas.io>



Let's Talk!



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